Lecture 02 – Review of Linear Regression

Stat 5376: Statistical Machine Learning

Ruiqi Liu

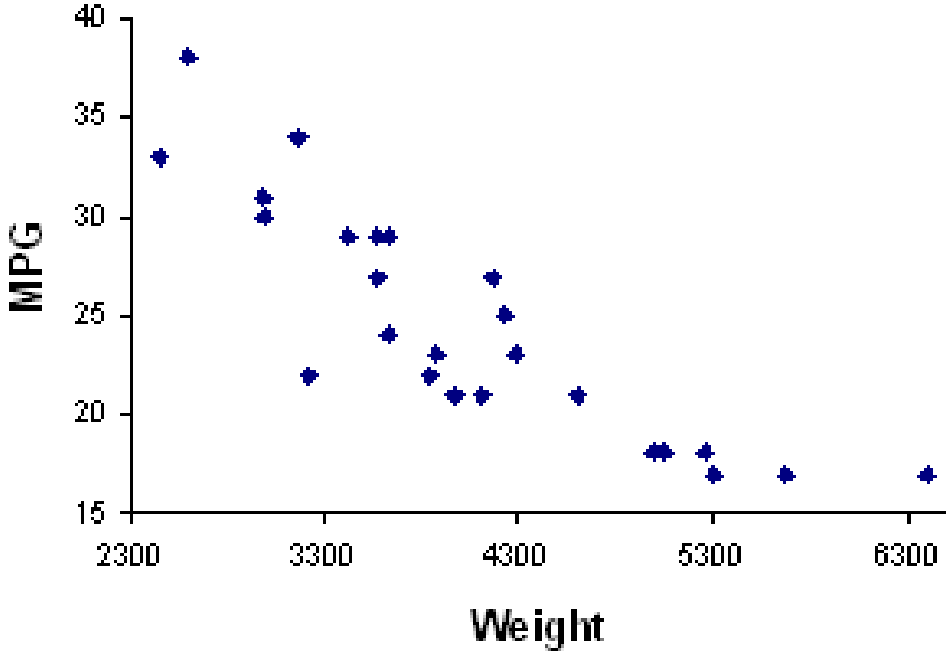
Department of Mathematics and Statistics
Texas Tech University

http://www.math.ttu.edu/~ruiqliu/

Disclaimer: These notes have not been subjected to the usual scrutiny reserved for formal publications. They may be distributed outside this class only with the permission of the Instructor.

Example 1: Car weight V.S. MPG

- What is the relationship between the weight of a vehicle and the number of miles it travels per gallon of gasoline? To answer this question, a random sample of 25 vehicles was taken. The weight (in pounds) and the miles per gallon (MPG) was recorded for each car.
- Variables:
  - Explanatory: Weight
  - Response: MPG



Observations

- There is a Negative, somewhat linear relationship between MPG and Weight.
- How to determine if a linear association between two variables is statistically significant?
- We begin by trying to find the straight line that best fits the data.
  - This technique is known as Simple Linear Regression.
  - It is an attempt to describe the relationship between the variables.
- The simplest model we can think of:

$$\text{MPG} \approx \beta_0 + \beta_1 \text{Weight}$$

- Why $\approx$? (because I know it is not a good enough model...)

Simple Linear Regression Model

The Simple Linear Regression Model is given as:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,\ i = 1, \ldots, n$$

- y_i: dependent variable or response variable.
- x_i: independent variable or explanatory variable.
- ε_i: the error term or noise term in order to explain the discrepancy.
  - Something in the response variable y_i that cannot be completely determined by the explanatory random variable x_i;
  - For example: for any car that has the same weight, the MPG cannot be completely the same due to
    - road conditions, driver's driving habits, gasoline qualities;
    - all these factors contribute to MPG, whose impact are sort of unpredictable, can be put into the error term ε_i;

Simple Linear Regression Model

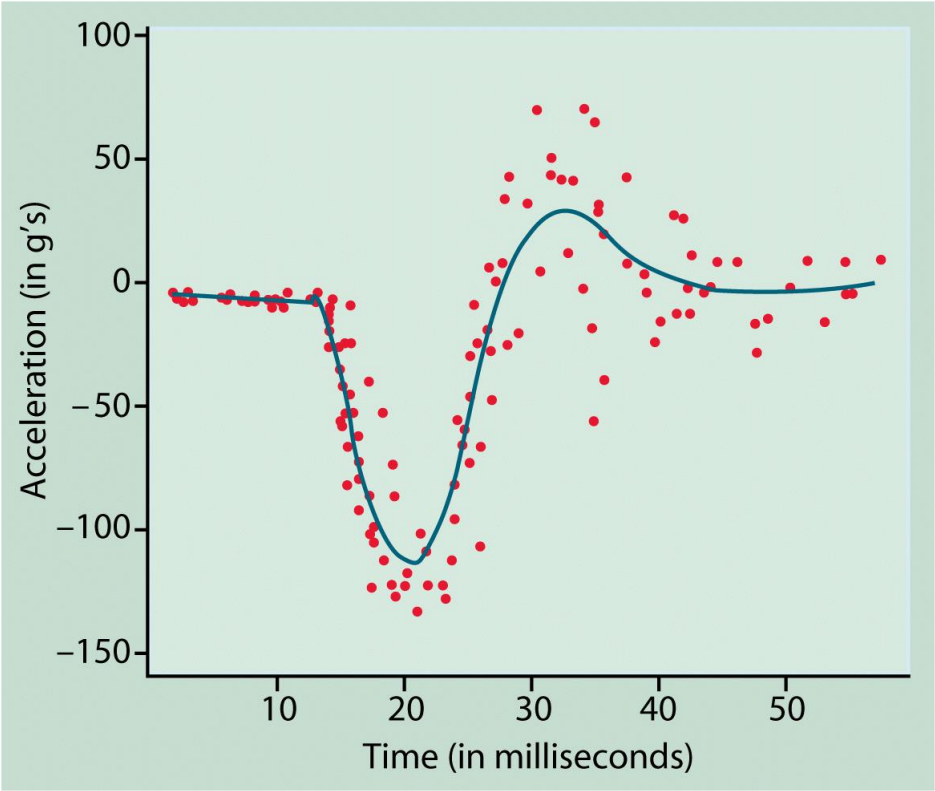
The Simple Linear Regression Model is given as:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,\ i = 1, \ldots, n$$

- β_0 is the intercept term and is fixed but unknown. (needs to be estimated!)
- β_1 is the slope term and is fixed but unknown. (needs to be estimated!)
- The explanatory random variable x_i is usually observable and thus considered fixed!
- ε_i's: independent, mean $E\varepsilon_i = 0$, constant variance $var(\varepsilon_i) = \sigma^2$.

Something to Keep in Mind

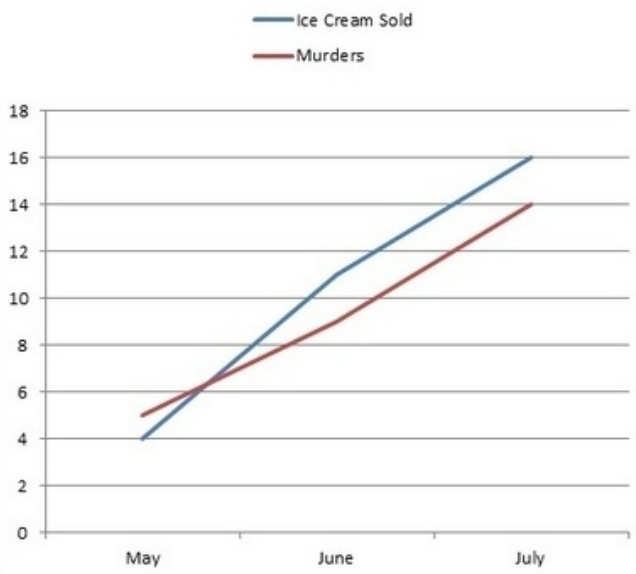
- Is the association between two variables really normal?
  - Not always true;
  - Nonlinear regression models are also available to deal with that.

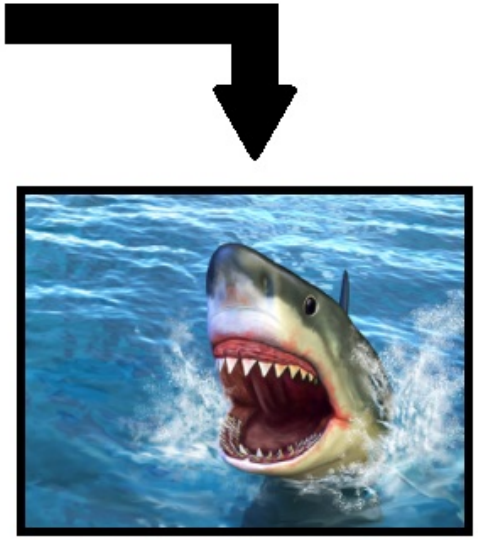


- Does the association really imply causation?
  - Not always true;
  - Florida, when ice cream sales goes up, the number of shark attacks goes up!
  - Definite association. But is there causation going on here?

Something to Keep in Mind

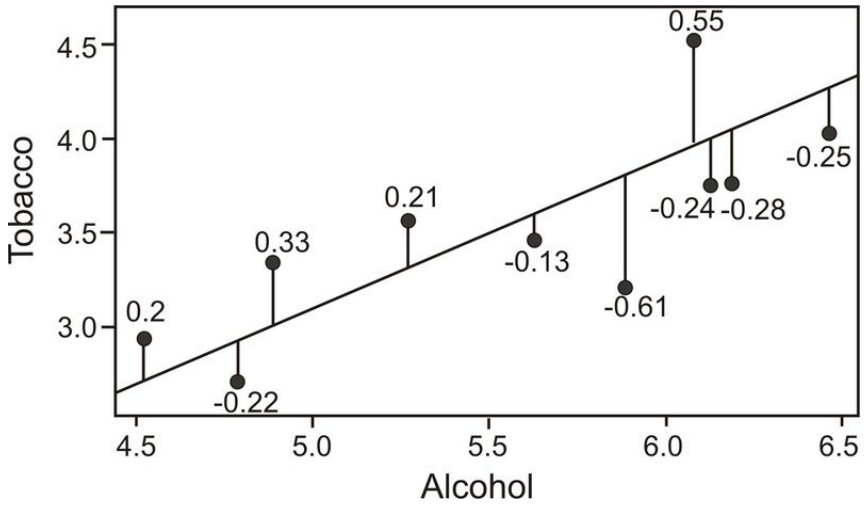
- Does the association really imply causation?
  - Not always true;
  - Florida, when ice cream sales goes up, the number of shark attacks also goes up!
  - Definite association. But is there causation going on here?





Least Square Estimator

- Method of Least Square
  - We attempt to find an estimator of β_0 and β_1, such that y_i's are overall "close" to the fitted line;
  - Define the fitted line as $\widehat{y}_i \equiv \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ and the residual, $e_i = y_i - \widehat{y}_i$;



  - We define the sum of squared errors (or residual sum of squares) to be

$$SSE(RSS) \equiv \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 = \sum_{i=1}^{n} (y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 x_i))^2$$

  - We **find a pair of $\widehat{\beta}_0$ and $\widehat{\beta}_1$ to minimize this sum of squares.** The resulting estimators are called least square estimators (LSE), or ordinary least square (OLS) estimators.

OLS Solution

- The OLS estimators for model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,\ i = 1, \ldots, n$$

are

$$\begin{cases} \widehat{\beta}_0 & = \overline{y} - \widehat{\beta}_1 \overline{x} \\ \widehat{\beta}_1 & = \frac{\sum_{i=1}^n (x_i - \overline{x}) y_i}{\sum_{i=1}^n (x_i - \overline{x})^2} \end{cases}$$

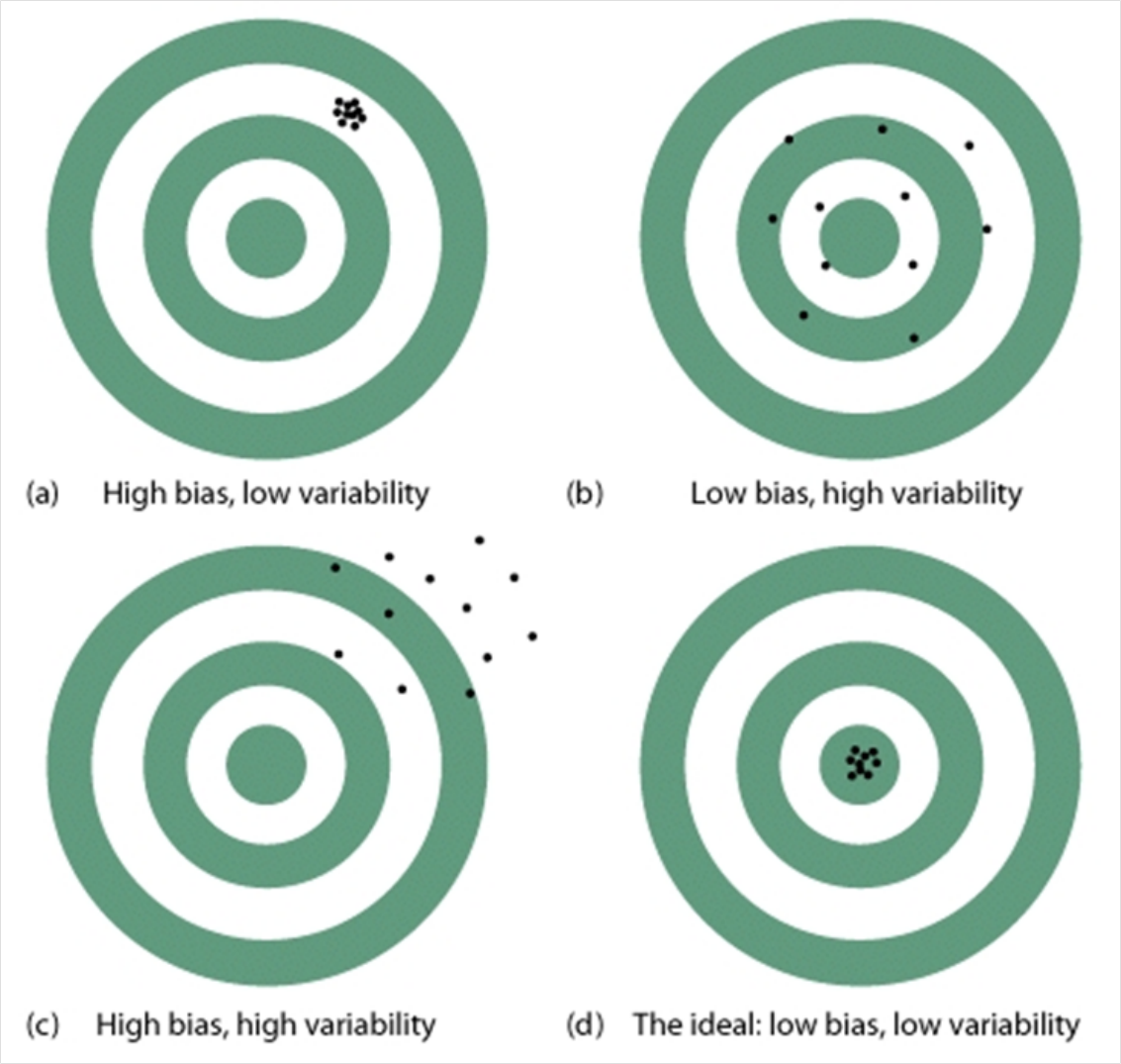
Notations

- Define $S_{xx} = \sum_{i=1}^{n}(x_i - \overline{x})^2$;
- Define $S_{xy} = \sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})$;
- We have the model set up (1):

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,\ i = 1, \ldots, n$$

  - Conditions: $\mathbb{E}\varepsilon_i = 0$ and $\mathrm{Var}\,\varepsilon_i^2 = \sigma^2$;
  - The Least square estimator are:
    - $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1\overline{x}$; $\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$;

Properties of Good Estimators



Unbiasedness

- Biases:
  - $\mathbb{E}[\widehat{\beta}_1] = \beta_1$.
  - $\mathbb{E}(\widehat{\beta}_0) = \beta_0$.
- Variances:
  - $\mathrm{Var}(\widehat{\beta}_1) = \mathrm{Var}\left[\sum_{i=1}^{n} \frac{(x_i - \overline{x})Y_i}{S_{xx}}\right] = \left[\frac{\left(\sum_{i=1}^{n}(x_i - \overline{x})^2\right)\sigma^2}{S_{xx}^2}\right] = \frac{\sigma^2}{S_{xx}}$
  - $\mathrm{Var}(\widehat{\beta}_0) = \sigma^2\left(\frac{\sum_{i=1}^{n} x_i^2}{nS_{xx}}\right)$
  - $Cov(\widehat{\beta}_0, \widehat{\beta}_1) = -\sigma^2\left(\frac{\overline{x}}{S_{xx}}\right)$

  Questions:
  - What does it mean $\mathrm{Var}(\widehat{\beta}_1) \neq 0$?
  - What is the source of randomness in $\widehat{\beta}_1$?
  - How can you reduce $\mathrm{Var}(\widehat{\beta}_1)$?

Visualization of Sampling Randomness

- You can copy the following code into your R program;
- Change the values of N, sigma, see what happens.

```
 N <- 30                                # Picking a sample size
 X <- cbind(x0=rep(1,N), x1=runif(N))  # Making up a predictor matrix
 sigma <- 1                  # Assuming a 'true' error variance
 beta <-c(b0=1, b1=2)   # Assuming a 'true' coeff vector
 y <- X%*%beta + rnorm(N,0, s=sigma)   # Simulating a response vector
 b <- solve(t(X)%*%X)%*%t(X)%*%y;  #ols estimator
##Simulate and animate parallel universes/possible worlds/...:
for(i in 1:50) {
 y <- rnorm(N, m=X%*%beta, s=sigma)    # Simulating a response vector
 b <- solve(t(X)%*%X) %*% t(X) %*% y
 plot(x=X[,2], y=y, pch=16, col="gray20", ylim=c(-2,6))
 abline(b, lwd=3, col="red")
 dev.flush()
 Sys.sleep(0.5)
 }
```

Covariance between $\widehat{\beta}_0$ and $\widehat{\beta}_1$

- You can copy the following code into your R program;
- Change the values of N, sigma, see what happens.

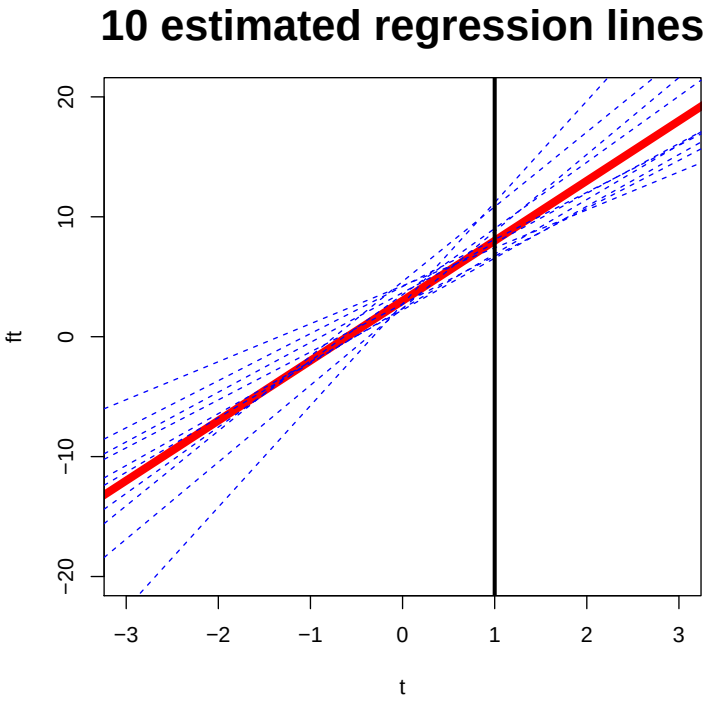
```
 N <- 30                                # Picking a sample size
 X <- cbind(x0=rep(1,N), x1=runif(N))  # Making up a predictor matrix
 sigma <- 1                  # Assuming a 'true' error variance
 beta <- c(b0=1, b1=2)   # Assuming a 'true' coeff vector
 ## Next, simulate many response vectors (possible worlds, parallel universes,...)
## and store the results:
 Nsim <- 10000
  bs <- matrix(NA, nrow=Nsim, ncol=length(beta))  # Storage for coeff estimates
  colnames(bs) <- paste("b",0:(length(beta)-1),sep="")  # Cosmetics
   for(isim in 1:Nsim) {
   y <- X%*%beta + rnorm(N,0, s=sigma) # Simulating a response vector
    bs[isim,] <- solve(t(X)%*%X)%*%t(X)%*%y
    if(isim%%100==0) cat(isim,"...")
    }
##Plot slopes versus intercepts to illustrate the 'sampling' or
## 'dataset-to-dataset' variability of b:
        plot(bs, pch=16, cex=.5)
##==> Negative correlation between b0 and b1 !!!
```

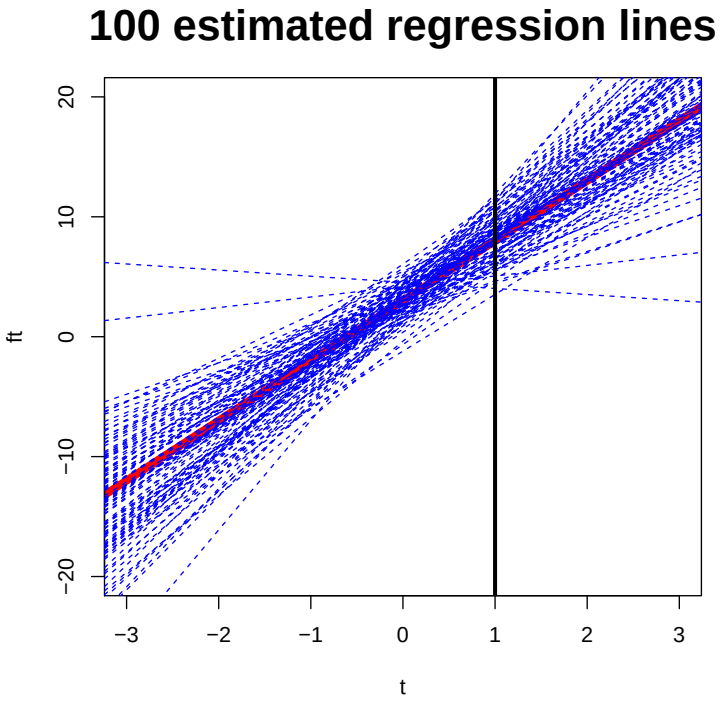
$\widehat{\beta}_1$ and $\widehat{\beta}_0$ Varies from Data to Data

- Let's simulate data from model

$$y_i = 3 + 5x_i + \epsilon_i, \quad i = 1, \ldots, n = 20.$$

- 100 different data sets were generated.





- Red line: when $x = 1$, $\mu_x = 3 + 5 \times 1 = 8$ <–the true mean value;
- Blue lines: when $x = 1$, the estimated value $\widehat{\mu}_x$ ranges from $[0, 10]$

Point Estimator V.S. Interval Estimator

- Point Estimator
  - A single number that is our “best guess" for the parameter of interest;
  - Examples: $\hat{\beta}_1$ for β_1 in a regression model;
  - Drawback: A point estimate by itself is not sufficient because we have no way of gauging the accuracy of the estimator (how close it is likely to be to the parameter).
- Interval Estimator: An interval of numbers within which the parameter value is believed to fall.
  - Example I: for β_1, $(\hat{\beta}_1 - b, \hat{\beta}_1 + b)$ is a set of possible values of β_1;
  - The length of the interval gauging the accuracy of the estimate.

Confidence Interval for β_1

- Interval estimators are commonly called *confidence interval.*
- Formed by the upper and lower confidence limits $(\hat{\beta}_L, \hat{\beta}_U)$;
- ***Confidence interval is random !***
- For different samples, the resulting confidence intervals are different. Some succeed in covering the true parameter; some fail.
- confidence coefficient $(1-\alpha)$: the probability that a (random) confidence interval $(\hat{\beta}_L, \hat{\beta}_U)$ covers the true parameter, that is

$$P(\hat{\beta}_L \leq \beta \leq \hat{\beta}_U) = 1 - \alpha$$

- $(\hat{\beta}_L, \hat{\beta}_U)$ provides a range of plausible values for true β;
- Then $(1-\alpha)100\%$ two-sided confidence interval for β_j is

$$[\hat{\beta}_1 - t_{1-\alpha/2} se(\hat{\beta}_j), \hat{\beta}_1 + t_{1-\alpha/2} se(\hat{\beta}_j)],$$

which has a $(1-\alpha)100\%$ chance of capturing the true β_1.

Hypothesis Testing

- Question: is there relationship between “MPG" and “Weight"?
- The most common hypothesis test involves testing the **null test hypothesis**

$$H_0: \text{ There is } \textbf{no} \text{ relationship between “MPG" and “Weight"}$$

versus the **alternative hypothesis**

$$H_1: \text{ There is } \textbf{some} \text{ relationship between “MPG" and “Weight"}$$

- When $\beta_1 = 0$, “MPG" does not change as “Weight" changes.
- Mathematically, this corresponds to testing

$$H_0 : \beta_1 = 0$$

versus

$$H_1 : \beta_1 \neq 0$$

Hypothesis Testing

- Hypothesis testing:
  - Set up $H_0 : \beta_1 = 0$ and $H_a : \beta_1 \neq 0$;
  - Find a point estimator for β_1: say $\widehat{\beta}_1$;
  - Find the standard error for $\widehat{\beta}_1$: $se(\widehat{\beta}_1)$;
  - Use the test statistic:

$$TS = \frac{\widehat{\beta}}{se(\widehat{\beta}_1)} \sim t_{df=n-2} \text{ assuming } H_0 \text{ is true, i.e } \beta_1 = 0;$$

  - Reject H_0 if TS is large; Fail to reject if TS is small.
  - Or, we can compute the p-value (by software),

$$\text{Reject } H_0 \text{ if p-value} < \alpha; \quad \text{Fail to reject } H_0 \text{ if p-value} \geq \alpha;$$

Evaluating the Fit

- After we fit the simple regression model, how can we know we fit a good model?
  - Use scatter plot to make sure the relationship between X and Y is linear;
  - Can we have a numerical measure?

ANOVA Decomposition

- Define notations:

$$TSS = \sum_{i=1}^{n} (y_i - \overline{y})^2, \qquad \text{Total sum of squares}$$

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2, \qquad \text{Residual sum of squares}$$

$$SS_{reg} = \sum_{i=1}^{n} (\hat{y}_i - \overline{y})^2, \qquad \text{Sum of squares in regression}$$

where $\hat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$;

- ANOVA Decomposition:

$$\underbrace{TSS}_{\text{Total variations in } Y} = \underbrace{SS_{reg}}_{\text{Variations in } Y \text{ explained by model}} + \underbrace{RSS}_{\text{Variations in } Y \text{ unexplained by model}}$$

Definition of R^2

The R^2 is defined as

$$R^2 = \frac{SS_{reg}}{TSS} = 1 - \frac{RSS}{TSS}$$

- R^2 is between 0 and 1;
- The bigger the R^2, the better (loosely speaking...);
- For simple linear regression, R^2 equals to r^2, the sample correlation between X and Y

$$R^2 = r^2 = \frac{\{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})\}^2}{\sum_{i=1}^n (x_i - \overline{x})^2 \sum_{i=1}^n (y_i - \overline{y})^2}.$$

Some Comments on R^2

- R^2 can be used to measure the strength of association;
- R^2 can only measure LINEAR relationship;
- If the relationship between x and y is not linear, R^2 is useless!
- R^2 and the scatter plot need to be used together! (The next four plots have exactly the same R^2!)

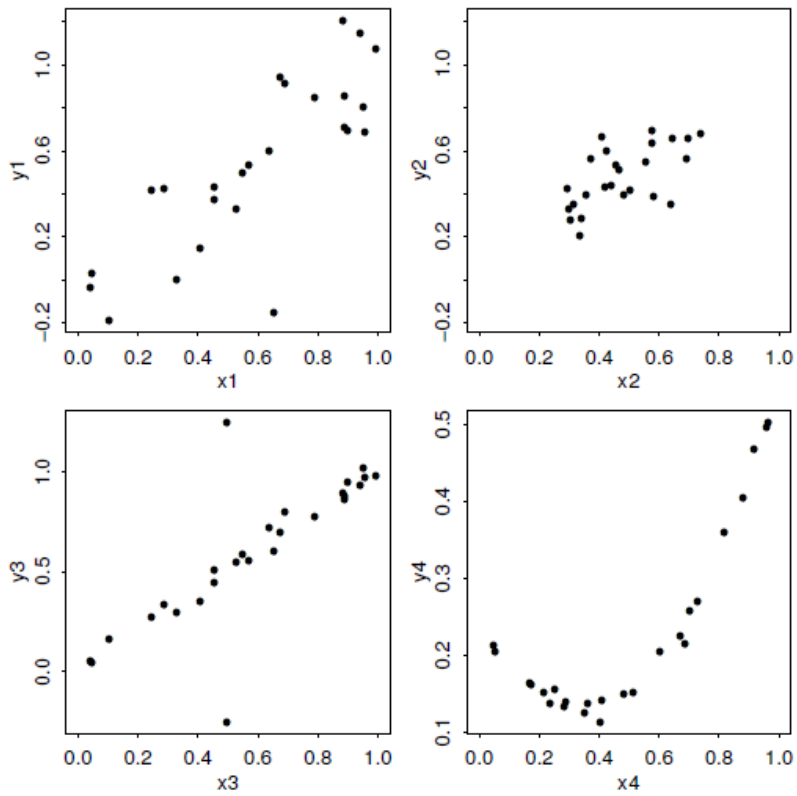


Figure 2.3 Four simulated datasets where R^2 is about 0.65. The plot on the upper

Multiple Linear Regression

- More than 1 independent variable; called multiple linear regression.

$$Y = \beta_0 + \beta_1 x_{(1)} + \beta_2 x_{(2)} + \cdots + \beta_p x_{(p)} + \varepsilon$$

  Notation: I deliberately use $x_{(j)}$ (subscript with parenthesis) to denote the jth variable, instead of the jth observation.

- Observations: $(y_i, x_{i1}, x_{i2}, \ldots, x_{ip})$, $i = 1, \ldots, n$.

$$\begin{aligned}
y_1 \leftarrow Y_1 &= \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_p x_{1p} + \varepsilon_1 \\
y_2 \leftarrow Y_2 &= \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \cdots + \beta_p x_{2p} + \varepsilon_2 \\
&\vdots \\
y_i \leftarrow Y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i \\
&\vdots \\
y_n \leftarrow Y_n &= \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \cdots + \beta_p x_{np} + \varepsilon_n
\end{aligned}$$

- We have $(p+1)$ coefficients (1 for the intercept term and p for the independent variables), and n observations. Each observation is represented by y_i and $\boldsymbol{x}_i := (1, x_{i1}, x_{i2}, \ldots, x_{ip})^T$

Matrix Representation

- Use matrix notation to ease the derivation:

$$\boldsymbol{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \underbrace{\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}}_{p+1 \text{ columns}} = \begin{bmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \\ \vdots \\ \boldsymbol{x}_n^T \end{bmatrix}$$

$$\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)^T, \quad \boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)^T$$

Rewrite model as

$$\boldsymbol{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Matrix Calculus/Gradient

For a scaler function $f(x_1, \ldots, x_k) = f(\boldsymbol{x}) \in \mathbb{R}$ and $\boldsymbol{x} \in \mathbb{R}^k$ is a vector. The gradient vector is defined as the vector of the partial derivatives of f with respect to x_i's, i.e.

$$\nabla f(\boldsymbol{x}) = \frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \frac{\partial f(\boldsymbol{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_k} \end{bmatrix}$$

Two examples:

- $\dfrac{\partial \boldsymbol{x}^T \boldsymbol{x}}{\partial \boldsymbol{x}} = \dfrac{\partial (\sum_{j=1}^k x_j^2)}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial \sum_{j=1}^k x_j^2}{\partial x_1} \\ \frac{\partial \sum_{j=1}^k x_j^2}{\partial x_2} \\ \vdots \\ \frac{\partial \sum_{j=1}^k x_j^2}{\partial x_k} \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \\ \vdots \\ 2x_k \end{bmatrix} = 2\boldsymbol{x}$
- $\dfrac{\partial \mathbf{A}\boldsymbol{x}}{\partial \boldsymbol{x}} = \mathbf{A}$ where $\mathbf{A}$ is a matrix that does not depend on $\boldsymbol{x}$.

- The least square estimator is obtained by minimizing the Residual Sum of Squares (RSS)

$$
\begin{aligned}
RSS(\widehat{\boldsymbol{\beta}}) &= \sum_{i=1}^{n} \{y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 x_{i1} + \widehat{\beta}_2 x_{i2} + \cdots + \widehat{\beta}_p x_{ip})\}^2 \\
&= \sum_{i=1}^{n} \{y_i - \boldsymbol{x}_i^T \widehat{\boldsymbol{\beta}}\}^2 \\
&= [\boldsymbol{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}]^T [\boldsymbol{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}]
\end{aligned}
$$

- To minimize $RSS(\widehat{\boldsymbol{\beta}})$, we have

$$
\mathbf{0} = \frac{\partial RSS(\widehat{\boldsymbol{\beta}})}{\partial \widehat{\boldsymbol{\beta}}} = -2\mathbf{X}^T[\boldsymbol{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}] = -2\mathbf{X}^T\boldsymbol{Y} + 2\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}}
$$

$$
\Rightarrow \mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\boldsymbol{Y} \Rightarrow \widehat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{Y}
$$

$\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\boldsymbol{Y}$ is called the estimating equation.

More Results from Multivariate Distribution

- If $\boldsymbol{Y} \in \mathbb{R}^p$ is a p-dimensional random vector, then its expectation is defined as the vector whose jth element is the expected value of the jth coordinate of $\boldsymbol{Y}$, i.e.

$$\mathbb{E}(\boldsymbol{Y}) = \begin{bmatrix} \mathbb{E}(Y_1) \\ \mathbb{E}(Y_2) \\ \vdots \\ \mathbb{E}(Y_n) \end{bmatrix}$$

- If $\mathbf{A}$ is a $m \times p$ nonrandom matrix, then $\mathbf{A}\boldsymbol{Y}$ is a m-dimensional random vector, whose expectation is

$$\mathbb{E}(\mathbf{A}\boldsymbol{Y}) = \mathbf{A}\mathbb{E}(\boldsymbol{Y})$$

More Results from Multivariate Distribution

- If $\boldsymbol{Y} \in \mathbb{R}^p$ is a p-dimensional random vector, then its (variance-)covariance matrix is defined as the $p \times p$ matrix whose ijth element is the covariance between the ith and jth coordinates of $\boldsymbol{Y}$, i.e.

$$\mathrm{Cov}(\boldsymbol{Y}) = \begin{bmatrix} \mathrm{Cov}(Y_1, Y_1) & \mathrm{Cov}(Y_1, Y_2) & \dots & \mathrm{Cov}(Y_1, Y_p) \\ \mathrm{Cov}(Y_2, Y_1) & \mathrm{Cov}(Y_2, Y_2) & \dots & \mathrm{Cov}(Y_2, Y_p) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(Y_p, X_1) & \mathrm{Cov}(Y_p, Y_2) & \dots & \mathrm{Cov}(Y_p, Y_p) \end{bmatrix}$$

- For example, $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ where $\mathbf{I}$ is the identity matrix which is a diagonal matrix with 1's on the diagonal.
- If $\mathbf{A}$ is a $m \times p$ nonrandom matrix, then the variance-covariance matrix of $\mathbf{A}\boldsymbol{Y}$ is

$$\mathrm{Cov}(\mathbf{A}\boldsymbol{Y}) = \mathbf{A}\,\mathrm{Cov}(\boldsymbol{Y})\mathbf{A}^T$$

Properties of $\widehat{\boldsymbol{\beta}}_{OLS}$

- Treat $\mathbf{X}$ as "nonrandom" quantity;
- $\widehat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{Y}$ is a $(p+1)$-vector, whose first element is the estimator for β_0 and whose $(j+1)$th element is the estimator for β_j, for $j = 1, \ldots, p$.
- $\mathbb{E}\{\widehat{\boldsymbol{\beta}}_{OLS}\} = \mathbb{E}\{(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{Y}\} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}\{\boldsymbol{Y}\}$. But recall that $\mathbb{E}\boldsymbol{Y} = \mathbf{X}\boldsymbol{\beta}$. Hence $\mathbb{E}\{\widehat{\boldsymbol{\beta}}_{OLS}\} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$. I.e. unbiasedness.
- And the covariance matrix is

$$\mathrm{Cov}(\widehat{\boldsymbol{\beta}}_{OLS}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

Inference on β_j

- Then $(1-\alpha)100\%$ two-sided confidence interval for β_j is

$$[\hat{\beta}_j - t_{1-\alpha/2} se(\hat{\beta}_j), \hat{\beta}_j + t_{1-\alpha/2} se(\hat{\beta}_j)],$$

  which has a $(1-\alpha)100\%$ chance of capturing the true β_j.
- Hypothesis test for $H_0 : \beta_j = 0$ v.s. $H_0 : \beta_j \neq 0$
  - Use the test statistic

$$T_j = \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} = \frac{\hat{\beta}_j - \beta_j}{s\sqrt{(\mathbf{X}^T\mathbf{X})^{-1}_{jj}}} \sim t_{df=n-p-1}$$

    where $(\mathbf{X}^T\mathbf{X})^{-1}_{jj}$ is the diagonal element in $(\mathbf{X}^T\mathbf{X})^{-1}$ that corresponds to jth predictor, and $s^2 = RSS(\widehat{\boldsymbol{\beta}})/(n-p-1)$.
  - we can compute the p-value (by software),

Reject H_0 if p-value$< \alpha$; Fail to reject H_0 if p-value$\geq \alpha$;

Example: Gala Data

```
>library(faraway)
###First check out the data
?gala
########
> lmod<-lm(Species ~ Area+Elevation+Nearest+Scruz+Adjacent,gala)
> sumary(lmod)
              Estimate Std. Error t value  Pr(>|t|)
(Intercept)  7.068221  19.154198  0.3690 0.7153508
Area        -0.023938   0.022422 -1.0676 0.2963180
Elevation    0.319465   0.053663  5.9532 3.823e-06
Nearest      0.009144   1.054136  0.0087 0.9931506
Scruz       -0.240524   0.215402 -1.1166 0.2752082
Adjacent    -0.074805   0.017700 -4.2262 0.0002971
n = 30, p = 6, Residual SE = 60.97519, R-Squared = 0.77
```

Then the 95% confidence interval for $\beta_{Adjacent}$ is

```
>  qt(0.975, 30-6)
[1] 2.063899
>  -0.074805 + c(-1,1) * 2.0639 *  0.017700
[1] -0.11133603 -0.03827397
```

Definition of R^2 for Multiple Regression

The R^2 is also defined as

$$R^2 = \frac{SS_{reg}}{TSS} = 1 - \frac{RSS}{TSS}$$

- R^2 is between 0 and 1 (if an intercept term is included!);
- The bigger the R^2, the better (loosely speaking...)

Lemma 1

It is easy to show that

$$R^2 = cor^2(\hat{\mathbf{Y}}, \mathbf{Y}),$$

the mean adjusted correlation between $\hat{\mathbf{Y}}$ *and* $\mathbf{Y}$.

Estimating σ^2

- So $\hat{\sigma}^2 = \frac{RSS}{n-p-1}$ is an unbiased estimator for σ^2. Recall that,
  - in simple linear regression, $p = 1$ and $\hat{\sigma}^2 = \frac{RSS}{n-2}$ was the unbiased estimator;
  - in iid case, $p = 0$ and $\hat{\sigma}^2 = \frac{RSS}{n-1}$ was the unbiased estimator.
    - in this case $RSS = \sum_{i=1}^{n}(y_i - \overline{y})^2$ and hence $\hat{\sigma}^2 = \frac{RSS}{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})^2$ is the sample variance.
  - $df_{reg} = p + 1$ is the degrees of freedom of the multiple regression model.

Linear Algebra Review

There is just one critical topic that we should address here, and that is linear dependence.

- The columns of the matrix A is linearly dependent

$$A = \begin{bmatrix} 1 & 2 & 4 & 1 \\ 2 & 1 & 8 & 6 \\ 3 & 6 & 12 & 3 \end{bmatrix}$$

  note that the third column is $4\times$ the first column;

- If none of the columns can be written as a linear combination of the other columns, then we say the columns are linearly independent.
- Unfortunately, linear dependence is not always obvious. For example, the columns in the following matrix A are linearly dependent

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 3 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

  note that the first column plus the second column equals $5\times$ the third column.

What is Multicollinearity?

- Multicollinearity exists whenever two or more of the predictors in a regression model are moderately or highly correlated.
- When this happens, it means some columns (2 or more) in the design matrix $\mathbf{X}$ are nearly linearly dependent.
- When the multicollinearity happens, the variance of the least square estimator $\widehat{\boldsymbol{\beta}}_{ols}$ maybe huge. Recall that

$$\mathrm{Var}(\widehat{\boldsymbol{\beta}}_{ols}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{|\mathbf{X}^T\mathbf{X}|} \times \underbrace{\mathbf{D}^*}_{\text{some matrix}} .$$

- When some columns of $\mathbf{X}$ are highly correlated (mathematically, nearly linearly dependent), the determinent

$$|\mathbf{X}^T\mathbf{X}| \approx 0,$$

- Therefore, $\mathrm{Var}(\widehat{\boldsymbol{\beta}}_{ols})$ will inflate! (What's the downside of this? Length of confidence interval will be huge, less useful!)
- We must deal with Multicollinearity!

Detecting Multicollinearity

- The first important thing to do is to detecting Multicollinearity;
- Recall that the root cause of Multicollinearity is that some predictors in the regression are highly correlated!
- Why don't we study the pairwise correlation of all predictors?

Blood Pressure Data

- Download the data set from the Blackboard;
  - blood pressure (y = BP, in mm Hg)
  - age (x_1 = Age, in years)
  - weight (x_2 = Weight, in kg)
  - body surface area (x_3 = BSA, in sq m)
  - duration of hypertension (x_4 = Dur, in years)
  - basal pulse (x_5 = Pulse, in beats per minute)
  - stress index (x_6 = Stress)
- The researchers were interested in determining if a relationship exists between blood pressure and age, weight, body surface area, duration, pulse rate and/or stress level.

Pairwise Scatter Plots

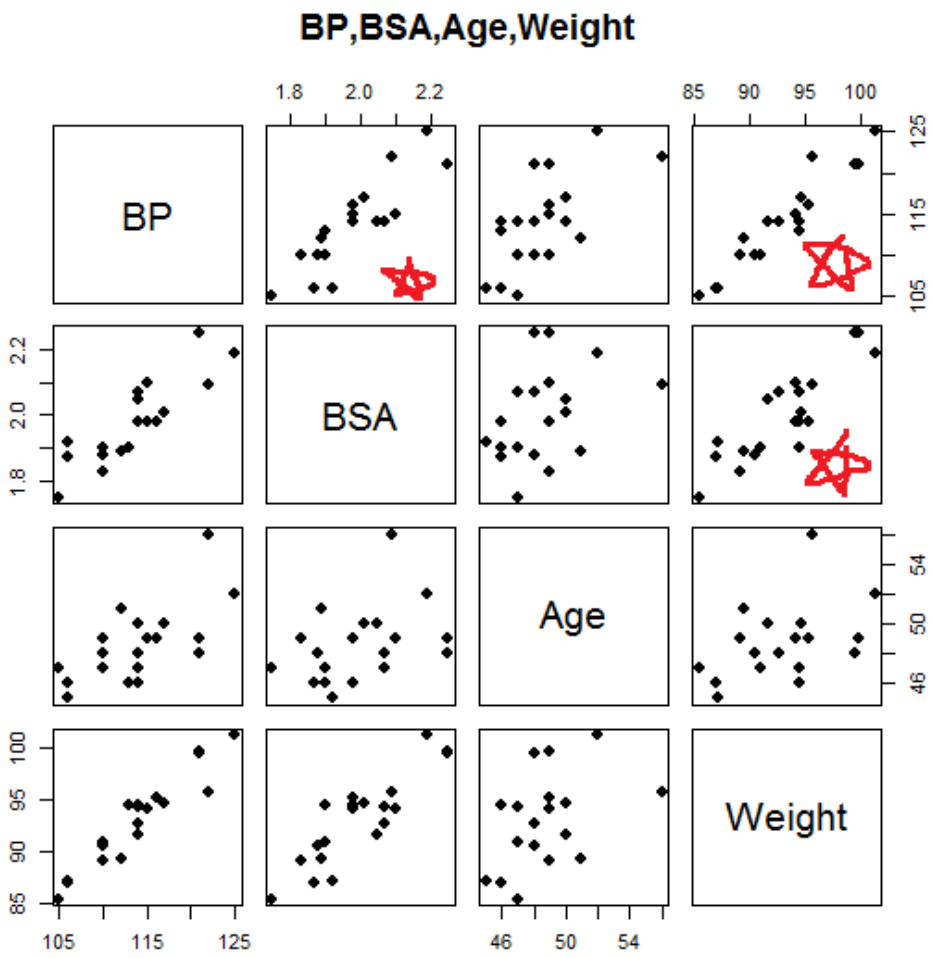
```
setwd("C:/study")
data <- read.table("bloodpress.txt",header=TRUE)

# Basic scatterplot matrix of the four measurements-first group
pairs(~BP+BSA+Age+Weight, data=data,pch=16, cex=1.2,main="BP,BSA,Age,Weight")

# Basic scatterplot matrix of the four measurements-second group
pairs(~BP+Dur+Pulse+Stress,data=data,pch=16,cex=1.2,main="BP,Dur,Pulse,Stress")

#####correlation matrix of all predictors###
round(cor(data),3) ###round() gives the first 3 decimal places
```

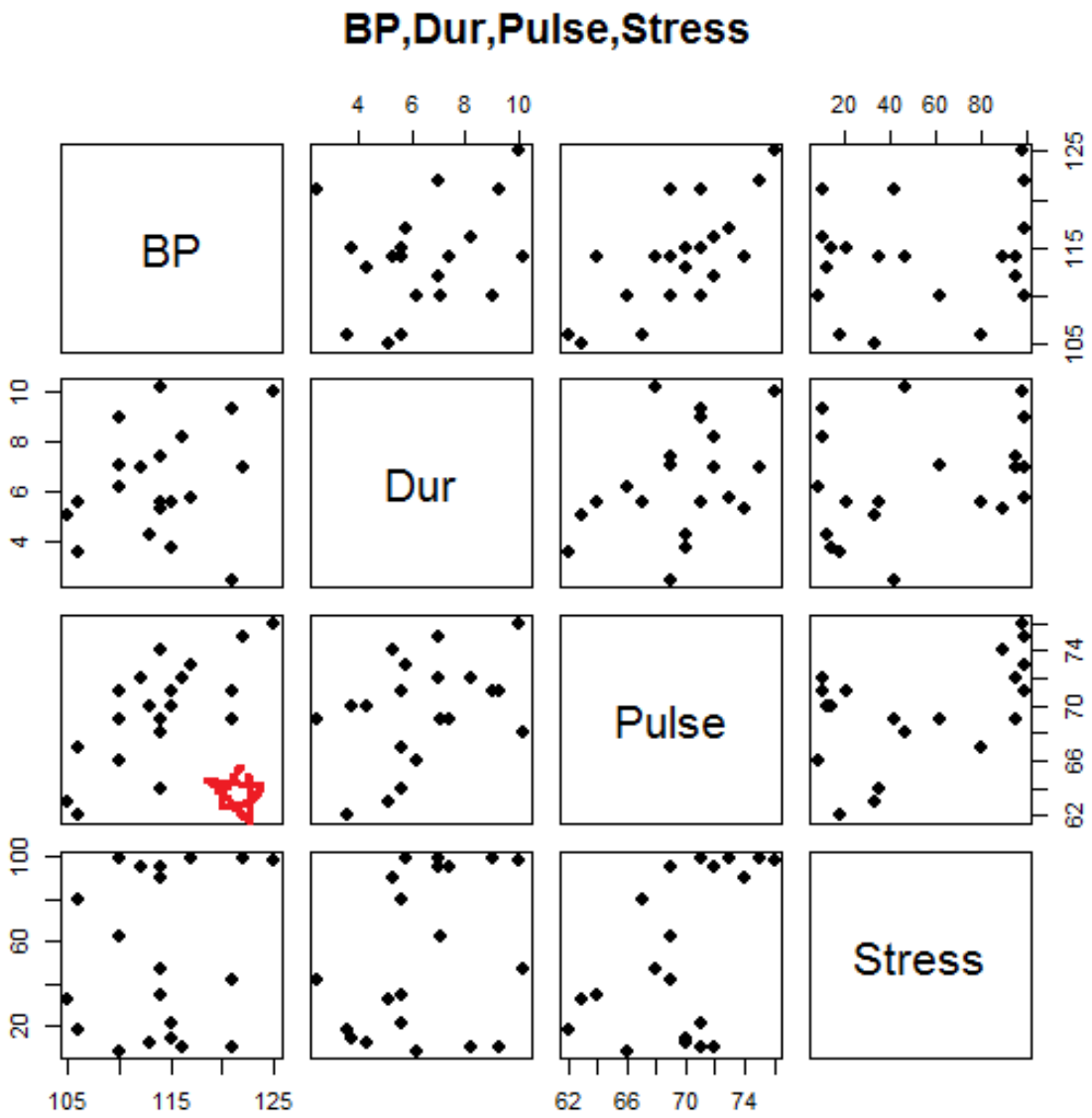
Pairwise Scatter Plots



- Weight and BSA appear to be strongly related;
- Stress and BSA appear to be hardly related at all;

Pairwise Scatter Plots

Do you see anything suspicious?



Correlation Matrix

```
round(cor(data),3)
```

	Pt	BP	Age	Weight	BSA	Dur	Pulse	Stress
Pt	1.000	0.031	0.043	0.025	-0.031	0.176	0.112	0.343
BP	0.031	1.000	0.659	0.950	0.866	0.293	0.721	0.164
Age	0.043	0.659	1.000	0.407	0.378	0.344	0.619	0.368
Weight	0.025	0.950	0.407	1.000	0.875	0.201	0.659	0.034
BSA	-0.031	0.866	0.378	0.875	1.000	0.131	0.465	0.018
Dur	0.176	0.293	0.344	0.201	0.131	1.000	0.402	0.312
Pulse	0.112	0.721	0.619	0.659	0.465	0.402	1.000	0.506
Stress	0.343	0.164	0.368	0.034	0.018	0.312	0.506	1.000

The high correlation among some of the predictors suggests that multicollinearity exists.

Use Only BAS to Predict BP

- Let the blood pressure (BP) be the response variable;
- Let BAS and Weight be two predictors;
- From above discussion, we know each one of them are good predictor for BP (because of high correlation)

```
setwd("C:/study")
data <- read.table("bloodpress.txt",header=TRUE)
#######Use only BSA as the predictor
obj1 <- lm(BP~BSA,data=data)

> summary(obj1)

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)   45.183      9.392   4.811  0.00014 ***
BSA           34.443      4.690   7.343 8.11e-07 ***
---
Residual standard error: 2.79 on 18 degrees of freedom
Multiple R-squared:  0.7497,    Adjusted R-squared:  0.7358
F-statistic: 53.93 on 1 and 18 DF,  p-value: 8.114e-07
```

Use Only Weight to Predict BP

```
setwd("C:/study")
data <- read.table("bloodpress.txt",header=TRUE)
#######Use only Weight as the predictor
obj2 <- lm(BP~Weight,data=data)

> summary(obj2)

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.20531    8.66333   0.255    0.802
Weight       1.20093    0.09297  12.917 1.53e-10 ***
---
Residual standard error: 1.74 on 18 degrees of freedom
Multiple R-squared:  0.9026,    Adjusted R-squared:  0.8972
F-statistic: 166.9 on 1 and 18 DF,  p-value: 1.528e-10
```

Use BAS and Weight to Predict BP

```
setwd("C:/study")
data <- read.table("bloodpress.txt",header=TRUE)
#######Use both BSA and Weight as the predictor
obj3 <- lm(BP~BSA+Weight,data=data)

> summary(obj3)

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.6534     9.3925   0.602    0.555
BSA           5.8313     6.0627   0.962    0.350
Weight        1.0387     0.1927   5.392 4.87e-05 ***
---

Residual standard error: 1.744 on 17 degrees of freedom
Multiple R-squared:  0.9077,    Adjusted R-squared:  0.8968
F-statistic: 83.54 on 2 and 17 DF,  p-value: 1.607e-09
```

Summary

Model	Coef. (BSA)	Std. (BSA)	Coef.(Weight)	Std. (Weight)
BSA ony	34.443	4.690	N/A	N/A
Weight only	N/A	N/A	1.20093	0.09297
BAS+Weight	5.8313 (insig)	6.0627	1.0387	0.1927

Intuitive explanation:

- Both BSA and Weight are good predictors for BP, individually;
- But BSA and Weight contains almost the same information (they are highly correlated);
- By adding Weight to the simple linear regression model (BP BSA), not much new information is added;
- However, by adding Weight to the simple linear regression model (BP BSA), the model becomes more complicated, (more things to estimate with the same amount of information) leads to bigger estimation error (reflected by the larger standard error.)
- In short (Good predictor+Good predictor $\not\Rightarrow$ Better model!). (which is bad...)

Variance Inflation Factors

To derive the “Variance Inflation Factor", we need to consider two models

- Model 1 (simple model): $y_i = \beta_0 + \beta_k x_{ki} + \varepsilon_i,$
- Under this model, the variance of least square model $\hat{\beta}_k$ is

$$\mathrm{Var}(\hat{\beta}_k)_{min} = \frac{\sigma^2}{\sum_{i=1}^n (x_{ki} - \bar{x}_k)^2}$$

- Model 2 (full model):

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \cdots + \beta_p x_{pi} + \varepsilon_i,$$

  where $x_1, \ldots, x_p$ are potentially correlated predictors;
- Under this model, the variance of least square model $\hat{\beta}_k$

$$\mathrm{Var}(\hat{\beta}_k) = \frac{\sigma^2}{\sum_{i=1}^n (x_{ki} - \bar{x}_k)^2} \times \frac{1}{1 - R_k^2} \qquad \text{(proof?)}$$

- R_k^2 is the R^2-value obtained by regressing the kth predictor x_k on the other predictors.

Variance Inflation Factors

- If some of the predictors are correlated with the predictor x_k, then the variance of $\hat{\beta}_k$ is inflated;
- R_k^2 can be obtained from the regression model

$$x_{ki} = \alpha_0 + \alpha_1 x_{1i} + \cdots + \alpha_{k-1} x_{(k-1)i} + \alpha_{k+1} x_{(k+1)i} + \cdots + \alpha_p x_{pi} + \epsilon_i;$$

- The Variance Inflation Factor is defined as

$$VIF_k = \frac{\mathrm{Var}(\hat{\beta}_k)}{\mathrm{Var}(\hat{\beta}_k)_{min}} = \frac{\left(\frac{\sigma^2}{\sum(x_{ik}-\bar{x}_k)^2} \times \frac{1}{1-R_k^2}\right)}{\left(\frac{\sigma^2}{\sum(x_{ik}-\bar{x}_k)^2}\right)} = \frac{1}{1-R_k^2}$$

- Recall that the interpretation of R_k^2: the percent of variation in x_k explained by other predictors;
- VIF_k measures how much the variance of the estimated regression coefficient $\hat{\beta}_k$ is “inflated" by the existence of correlation among the predictor variables in the model;
- $VIF_k = 1$: no inflation; $VIF_k \geq 4$: needs investigation; $VIF_k > 10$: serious multicollinearity requiring correction.

Blood Pressure Data

```
setwd("C:/study")
data <- read.table("bloodpress.txt",header=TRUE)
#######Fitting the full model of the blood pressure data
obj4 <- lm(BP~Age+Weight+BSA+Dur+Pulse+Stress,data=data)
>summary(obj4)
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.870476   2.556650  -5.034 0.000229 ***
Age           0.703259   0.049606  14.177 2.76e-09 ***
Weight        0.969920   0.063108  15.369 1.02e-09 ***
BSA           3.776491   1.580151   2.390 0.032694 *
Dur           0.068383   0.048441   1.412 0.181534
Pulse        -0.084485   0.051609  -1.637 0.125594
Stress        0.005572   0.003412   1.633 0.126491
---
Residual standard error: 0.4072 on 13 degrees of freedom
Multiple R-squared:  0.9962,    Adjusted R-squared:  0.9944
F-statistic: 560.6 on 6 and 13 DF,  p-value: 6.395e-15

> vif(obj4) # VIF of all predictors
Age   Weight      BSA      Dur    Pulse   Stress
1.762807 8.417035 5.328751 1.237309 4.413575 1.834845
```

Blood Pressure Data

- Let's verify the calculation of the VIF for the predictor Weight.

```
setwd("C:/study")
data <- read.table("bloodpress.txt",header=TRUE)
##Regressing the predictor Weight on the remaining five predictors:
obj5 <- lm(Weight~Age+BSA+Dur+Pulse+Stress,data=data)
>summary(obj5)
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 19.674438   9.464742   2.079  0.05651 .
Age         -0.144643   0.206491  -0.700  0.49510
BSA         21.421654   3.464586   6.183 2.38e-05 ***
Dur          0.008696   0.205134   0.042  0.96678
Pulse        0.557697   0.159853   3.489  0.00361 **
Stress      -0.022997   0.013079  -1.758  0.10052
---
Residual standard error: 1.725 on 14 degrees of freedom
Multiple R-squared:  0.8812,    Adjusted R-squared:  0.8388
F-statistic: 20.77 on 5 and 14 DF,  p-value: 5.046e-06
```

The $R^2_{weight} = 0.8812$ and thus

$$VIF_{weight} = \frac{1}{1 - R^2_{weight}} = \frac{1}{1 - 0.8812} = 8.417$$

How to Reduce Multicollinearity?

- We have learn how to detect multicollinearity.
- How can we reduce the multicollinearity in the regression model
  - The first thing we can try: remove the predictors with high VIF from the model. (Sometimes work!)
  - Ridge regression;
  - Partial Least Squares Regression;
  - Principal Components regression;
  - $\vdots$

Removing Weight from Model

- What would happen if we remove the predictor “Weight" from the model?

```
setwd("C:/study")
data <- read.table("bloodpress.txt",header=TRUE)
##Regressing the predictor Weight on the remaining five predictors:
obj6 <- lm(BP~Age+BSA+Dur+Pulse+Stress,data=data)
>summary(obj6)
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.21215    9.42927   0.659   0.5207
Age          0.56297    0.20572   2.737   0.0161 *
BSA         24.55378    3.45160   7.114 5.22e-06 ***
Dur          0.07682    0.20437   0.376   0.7126
Pulse        0.45644    0.15925   2.866   0.0124 *
Stress      -0.01673    0.01303  -1.284   0.2199
---
Residual standard error: 1.718 on 14 degrees of freedom
Multiple R-squared:  0.9262,    Adjusted R-squared:  0.8998
F-statistic: 35.14 on 5 and 14 DF,  p-value: 1.921e-07

> vif(obj6) # VIF of all predictors
Age      BSA      Dur    Pulse   Stress
1.703115 1.428349 1.237151 2.360939 1.502936

No more multicollinearity!
```

Example: Predicting Income

- Sometimes we are interested in the effect of a categorical/qualitative predictor
- For example: do men earn more than women?
  - Response variable: income;
  - One potential numerical predictor: education (# of years in school);
  - One potential categorical predictor: gender (male or female), can take two values;
  - Another potential categorical predictor: occupation type (Blue collar (bc), Professional (prof), White collar (wc)), can take three values;

Simplest Model

- Let Y: response variable; D: predictor for gender (male or female);
- D is a dummy variable:

$$D_i = \begin{cases} 1 & \text{for men} \\ 0 & \text{for women} \end{cases}$$

- A simple linear regression model is

$$Y_i = \beta_0 + \beta_1 D_i + \varepsilon_i \quad i = 1, \ldots, n;$$

- Suppose that there are n_1 mean and n_2 women, that is $n = n_1 + n_2$. Let $\overline{Y}_{.1}$ and $\overline{Y}_{.2}$ be averages for men and women group.

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \\ 1 & 1_{n_1} \\ \hline 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 0_{n_2} \end{pmatrix}, \mathbf{X}^T\mathbf{X} = \begin{pmatrix} n & n_1 \\ n_1 & n_1 \end{pmatrix}, \hat{\boldsymbol{\beta}}_{ols} = \begin{pmatrix} \overline{Y}_{.2} \\ \overline{Y}_{.1} - \overline{Y}_{.2} \end{pmatrix}$$

Interpretation of Simplest Model

- For men's group ($D_i = 1$):

$$Y_i = \beta_0 + \beta_1(D_i = 1) + \varepsilon_i = (\beta_0 + \beta_1) + \varepsilon_i;$$

- For women's group ($D_i = 0$):

$$Y_i = \beta_0 + \beta_1(D_i = 0) + \varepsilon_i = (\beta_0) + \varepsilon_i;$$

- What is the interpretation of β_0 and β_1?
  - β_0: population mean for women group;
  - β_1: the difference in population means (men-women);
- The least square estimator of β_0 and β_1?
  - $\hat{\beta}_0 = \overline{Y}_{\cdot 2}$: the sample mean for women group;
  - $\hat{\beta}_1 = \overline{Y}_{\cdot 1} - \overline{Y}_{\cdot 2}$: the difference in sample means (men-women);
- Testing for $\beta_1 = 0$ is equivalent to two sample t-test!

Second coding of Dummy Variables

- If we change the coding of D as follows:

$$D_i = \begin{cases} 1 & \text{for men} \\ -1 & \text{for women} \end{cases},$$

  what would happen? (Note: a dummy variable can only take two possible values!)
- For men's group ($D_i = 1$):

$$Y_i = \beta_0 + \beta_1(D_i = 1) + \varepsilon_i = (\beta_0 + \beta_1) + \varepsilon_i;$$

- For women's group ($D_i = -1$):

$$Y_i = \beta_0 + \beta_1(D_i = -1) + \varepsilon_i = (\beta_0 - \beta_1) + \varepsilon_i;$$

- What is the interpretation of β_0 and β_1?
  - a bit harder to interpret!
- Coding of the Dummy variables changes the meaning of the parameters! be careful.
- For prediction purposes, coding of the Dummy variables does NOT have any impact.

Polytomous Predictors

- Let Y: response variable; Predictor for occupation type (Blue collar (bc), Professional (prof), White collar (wc));
- We need to define two dummy variables D_1 and D_2:

Type	D_1	D_2
Blue collar	0	0
Professional	1	0
White collar	0	1

- A multiple linear regression model is

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \varepsilon_i \quad i = 1, \ldots, n;$$

- The design matrix becomes

$$\mathbf{x} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Polytomous Predictors

- Can we use three dummy variables D_1, D_2, D_3 instead?

Type	D_1	D_2	D_3
Blue collar	1	0	0
Professional	0	1	0
White collar	0	0	1

- A multiple linear regression model is

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \varepsilon_i \quad i = 1, \ldots, n;$$

- The design matrix becomes

$$\mathbf{x} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

- $\mathbf{X}$ is NOT full rank! (How to fix this?)
- Remedies? Remove the intercept β_0 in the model? what would happen then? (Interpretation of the model parameters will be completely changed!)

Categorical Predictors with m Levels

- If your predictor has m different levels and you want to use a regression model with an intercept

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \cdots + \beta_{m-1} D_{m-1i} + \varepsilon_i \quad i = 1, \ldots, n;$$

- We should use m-1 dummy variables $D_1, D_2, \ldots D_{m-1}$.

Type	D_1	D_2	$\ldots$	D_{m-1}
Leve 1	0	0	$\ldots$	0
Leve 2	1	0	$\ldots$	0
Leve 3	0	1	$\ldots$	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Leve m	0	0	$\ldots$	1

- Pay attention to interpretations of $\beta_0, \beta_1, \ldots, \beta_{m-1}$ in the model!

Mixture of Numerical and Categorical Predictors

- Most of times in practice, you will have a mixture of numerical and categorical predictors in your hand;
  - Response variable: income (Y);
  - One potential numerical predictor: education (# of years in school) (X);
  - One potential categorical predictor: occupation type (Blue collar (bc), Professional (prof), White collar (wc)), can take three values; two dummy variables D_1 and D_2 are needed

Type	D_1	D_2
Blue collar	0	0
Professional	1	0
White collar	0	1

- One additional question you need to consider: are there any interactions between this two variables (numerical v.s. categorical)?
- For example: White collar workers are more likely to receive more education, which tends to give bigger boost of income!

Common Slope Model – no interactions

Model: income (Y) v.s. education (X) and occupation types (D_1, D_2)

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_{1i} + \gamma_2 D_{2i} + \varepsilon_i, \quad i = 1, \ldots, n.$$

- For Blue collar group ($D_1 = D_2 = 0$):

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

- For Professional group ($D_1 = 1, D_2 = 0$):

$$Y_i = (\beta_0 + \gamma_1) + \beta_1 X_i + \varepsilon_i,$$

- For White collar group ($D_1 = 1, D_2 = 1$):

$$Y_i = (\beta_0 + \gamma_1 + \gamma_2) + \beta_1 X_i + \varepsilon_i,$$

- For all these three groups, each group has its own regression line; (if there is no interaction, they are parallel!)
- White don't we fit a regression line for each of these groups separately? (answer: it would be bad for the common slop β_1!)

Interaction Model

Model: income (Y) v.s. education (X) and occupation types (D_1, D_2)

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_{1i} + \gamma_2 D_{2i} + \delta_1(X_i D_{1i}) + \delta_2(X_i D_{2i}) + \varepsilon_i, \quad i = 1, \ldots, n.$$

- For Blue collar group ($D_1 = D_2 = 0$):
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$
- For Professional group ($D_1 = 1, D_2 = 0$):
$$Y_i = (\beta_0 + \gamma_1) + (\beta_1 + \delta_1) X_i + \varepsilon_i,$$
- For White collar group ($D_1 = 1, D_2 = 1$):
$$Y_i = (\beta_0 + \gamma_1 + \gamma_2) + (\beta_1 + \delta_1 + \delta_2) X_i + \varepsilon_i,$$
- For all these three groups, each group has its own regression line; (if there is interaction, they are NOT parallel!)
- In other words, the value of the categorical predictor also affects the association between Y(income) and X(education);
- For example, as a white collar worker, you income is not as much affected by your education as that is for a blue collar worker (or the other way around;)

Simple Linear Regression Model

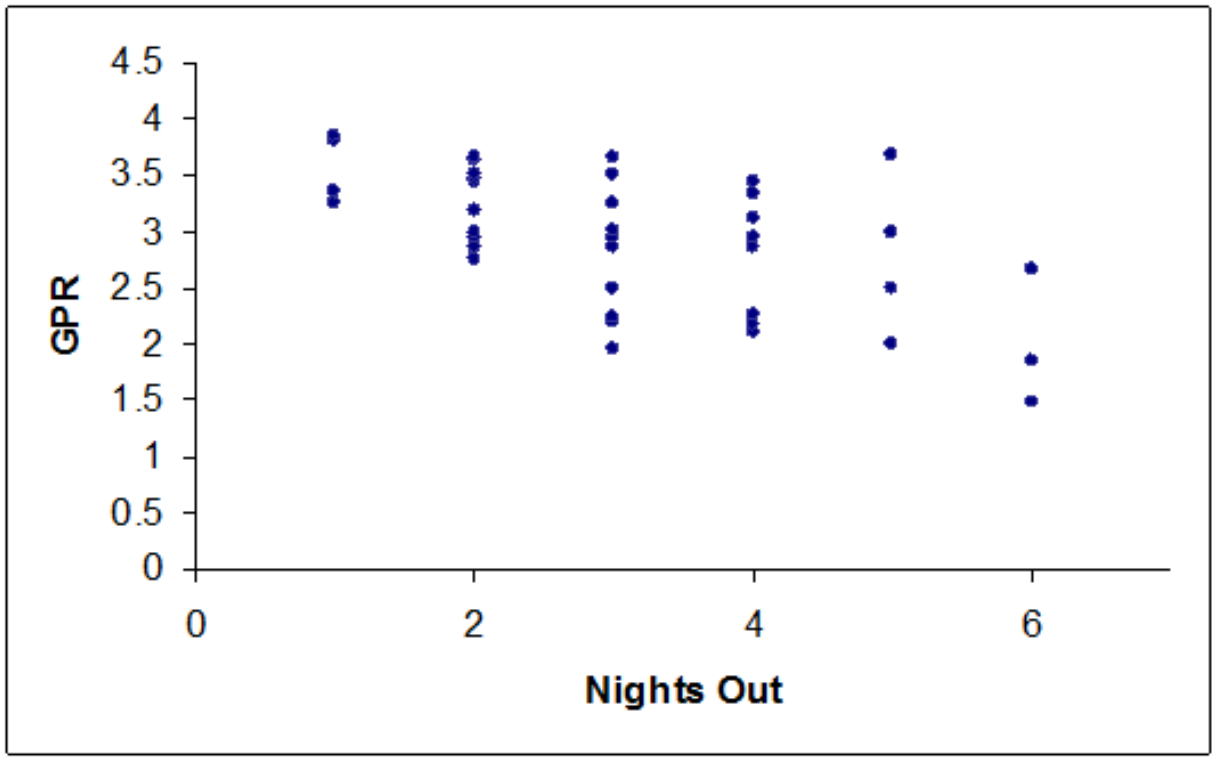
- We have the model set up (1):

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,\ i = 1, \ldots, n$$

- Conditions need to assume:
  - The type of relationship between the explanatory variable and response variable is Linear;
  - $\mathbb{E}\varepsilon_i = 0$ and $\mathrm{Var}\,\varepsilon_i^2 = \sigma^2$–Constant variance of error term ε_i's;
  - The error terms ε_i's has a normal distribution;
  - There is no outlier or influential point;
  - All error terms ε_i's are independent;
- Before you do the statistical analysis, you need to verify all these conditions to detect potential violations; but how?

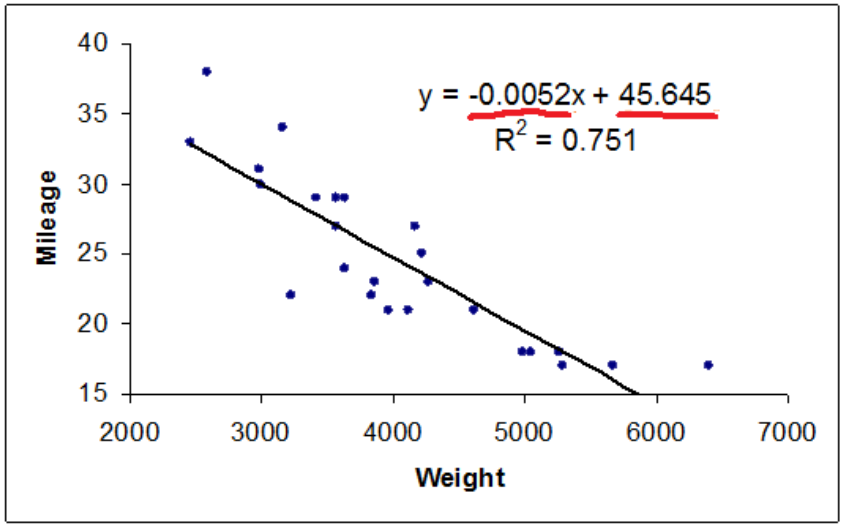
Detecting Linear Association

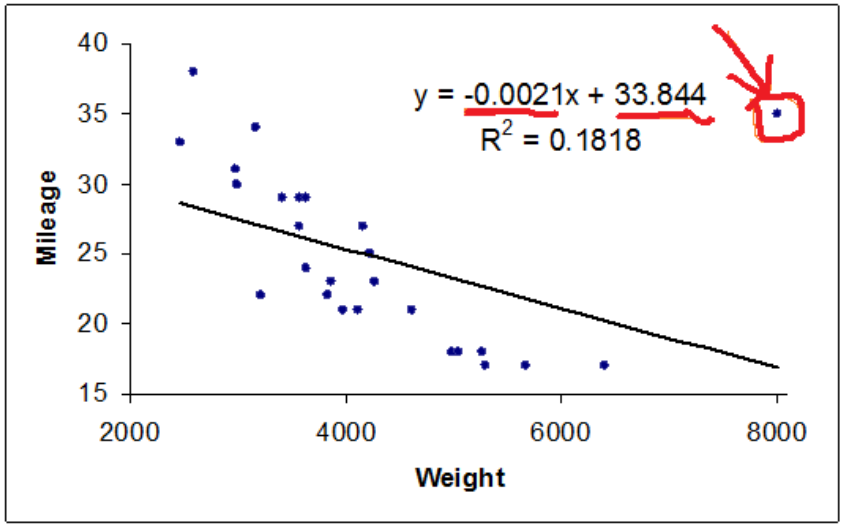
- Test the claim that the number of times a TTU student goes out per week is negatively related to their GPA.
- Explanatory variable: Nights out per week;
- Response variable: GPA
- Does there appear to be an association? If so, describe it.
  - There appears to be a weak negative association.



Detecting Influential Point/Outlier

- **An outlier** is an observation that does not follow the general pattern of the rest of the data;
  - It does not fall in the straight-line pattern with the rest of the data.
- Outliers can have a strong impact on the least square **estimators** of slope and intercept of the regression line.
- If your data has a potentially influential point, you should
  - Investigate the point¡ªis it possible that it was recorded incorrectly?
  - Refit the regression line without the point to see if it has a large effect.



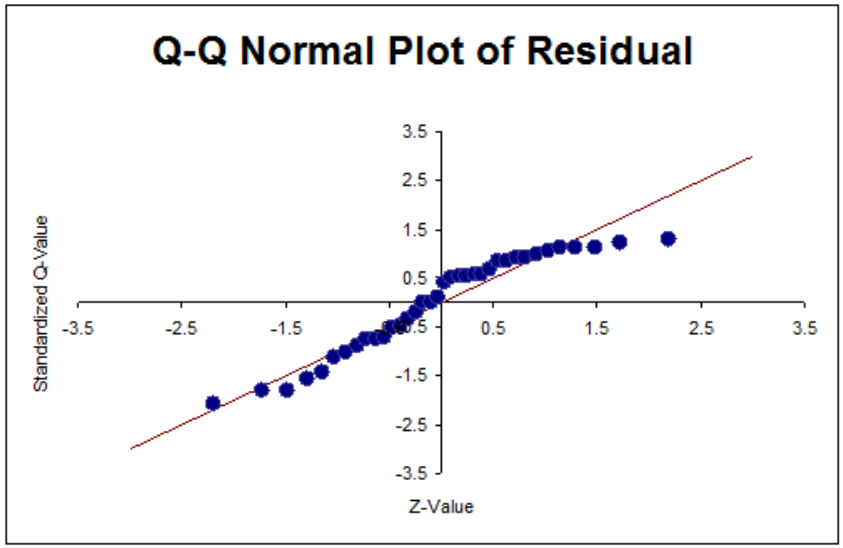


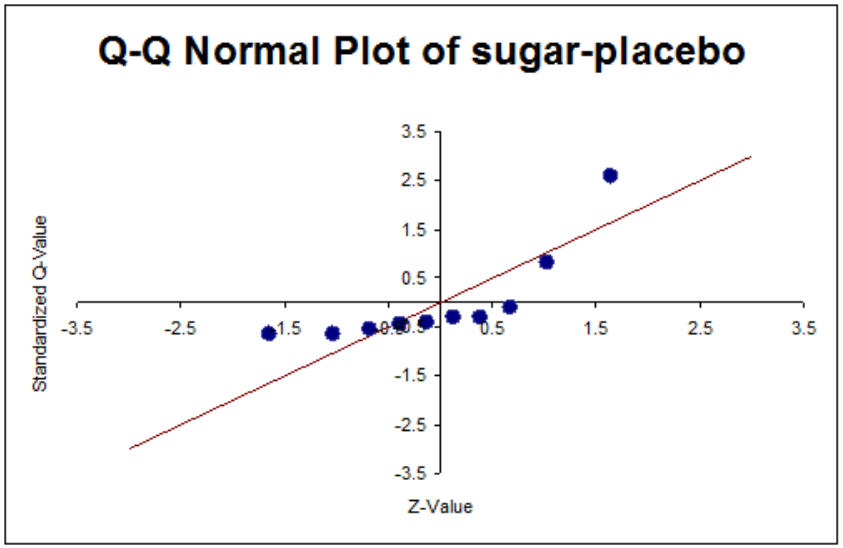
Checking normality

- After the regression line is fitted, calculate the residuals by

$$\hat{\varepsilon}_i = y_i - \widehat{y}_i$$

- Plot the Q-Q plot of the residuals $\hat{\varepsilon}$'s;
  - You can use R function *qqnorm()* to draw this plot;
- If at each x value, the distribution of the y's is normal, this plot should be a straight line. (Left pic: normal; right pic: non-normal)



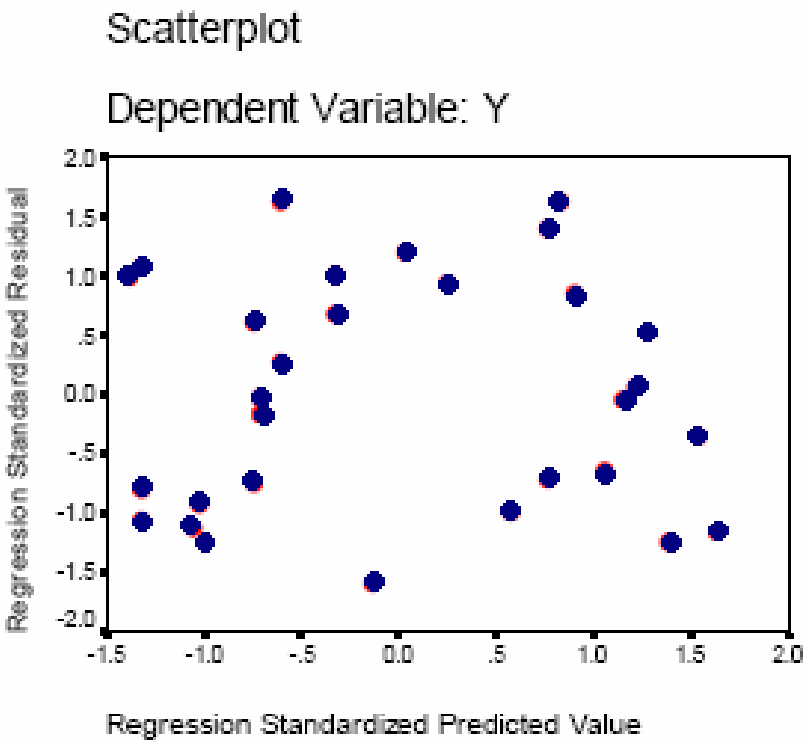


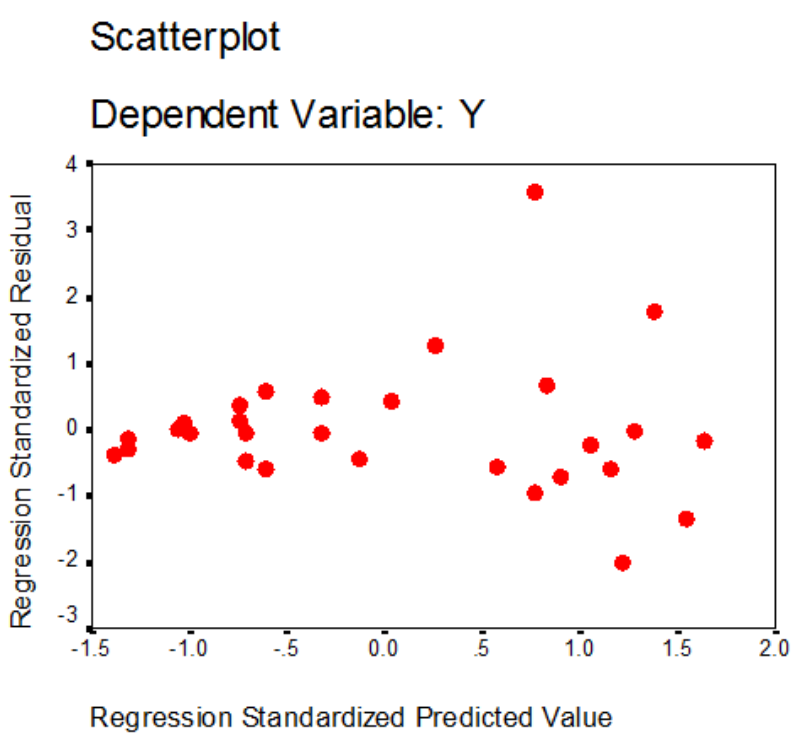
Checking Constant Variance

- After the regression line is fitted, calculate the residuals by

$$\hat{\varepsilon}_i = y_i - \widehat{y}_i$$

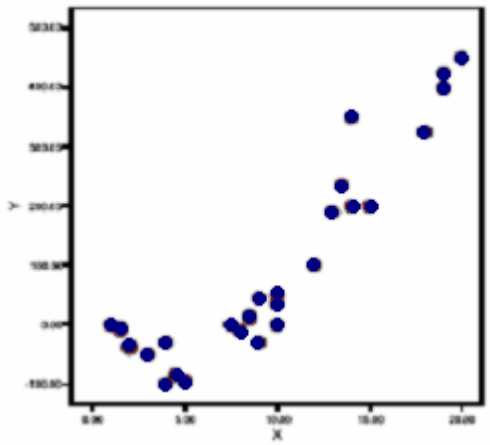
- Plot the residual plot of the residuals $\hat{\varepsilon}_i$'s: the predicted value $\widehat{y}_i$ as the x-axis and the residual e_i as the y-axis;
- But why do we choose to use the predicted value $\widehat{y}_i$ as the x-axis?
- If the constant variance assumption is met, the residual plot should look random; Left: look random; Right: there is a fan shape pattern:

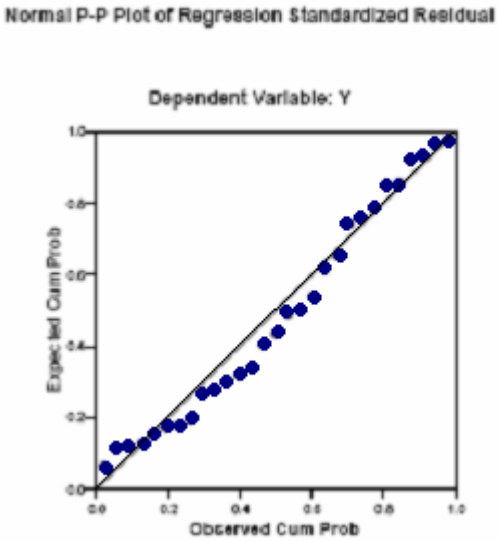


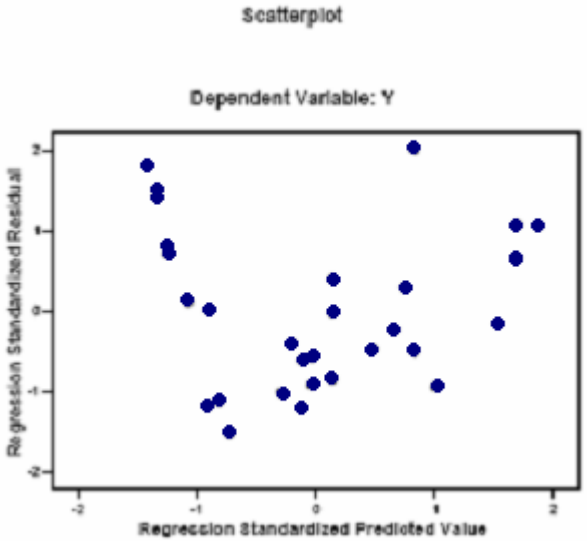


Potential problem # 1

- The relationship between x and y is not linear.

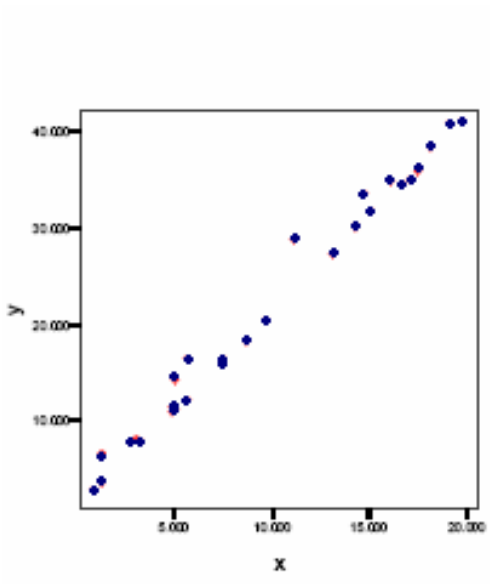


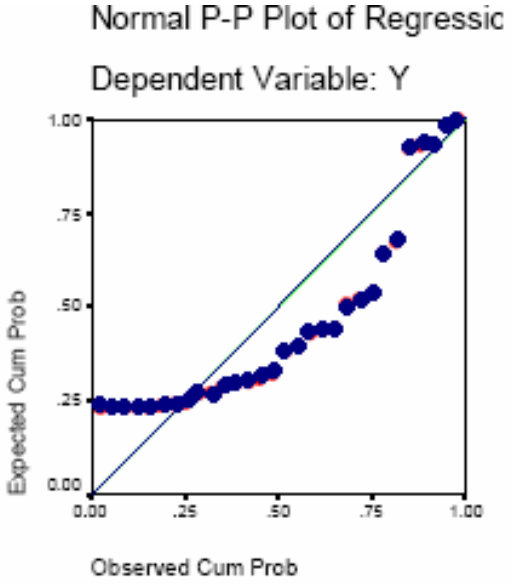


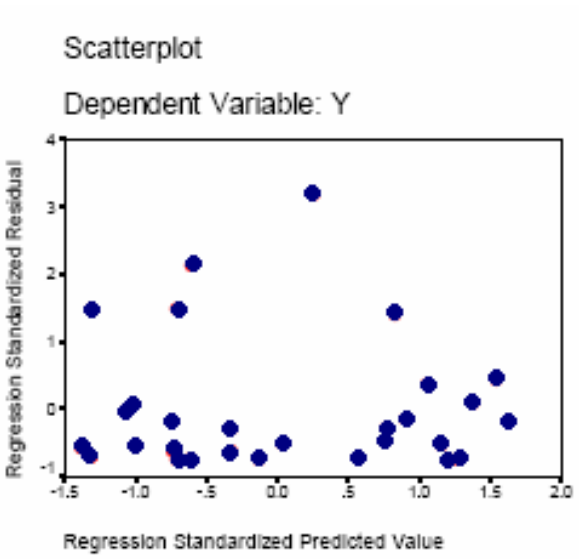


Potential problem # 2

- The distribution of the response variable (y) is not normal for each value of the explanatory variable (x) .

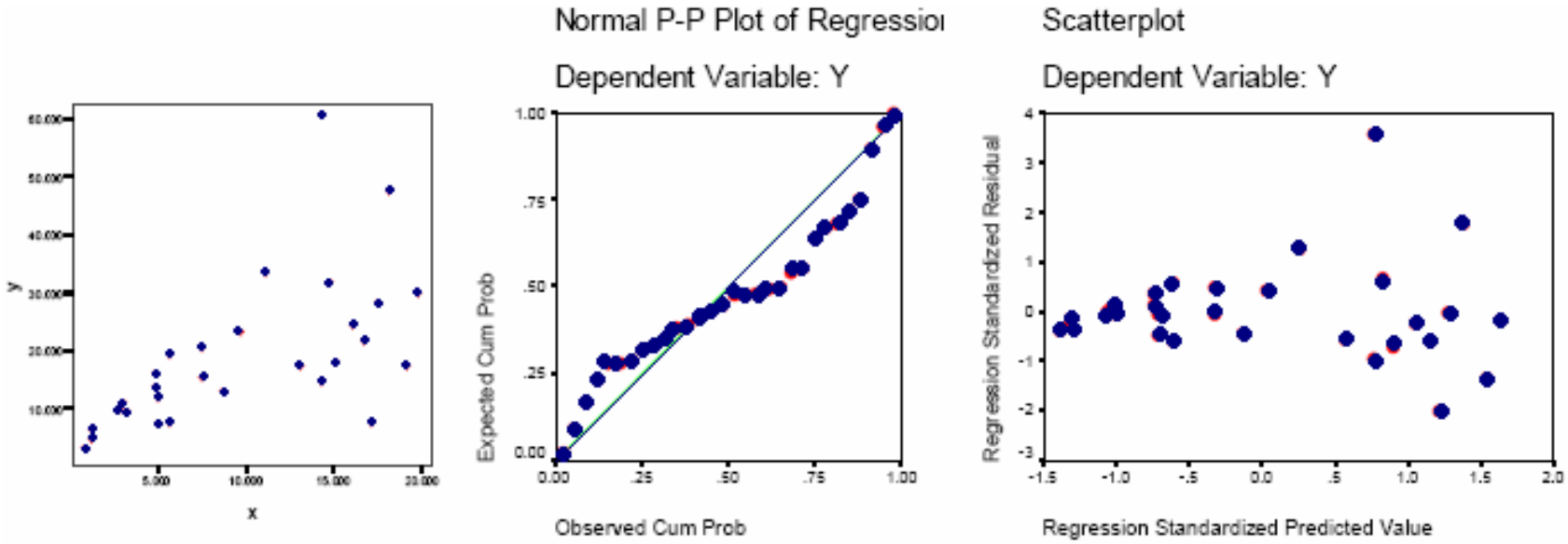






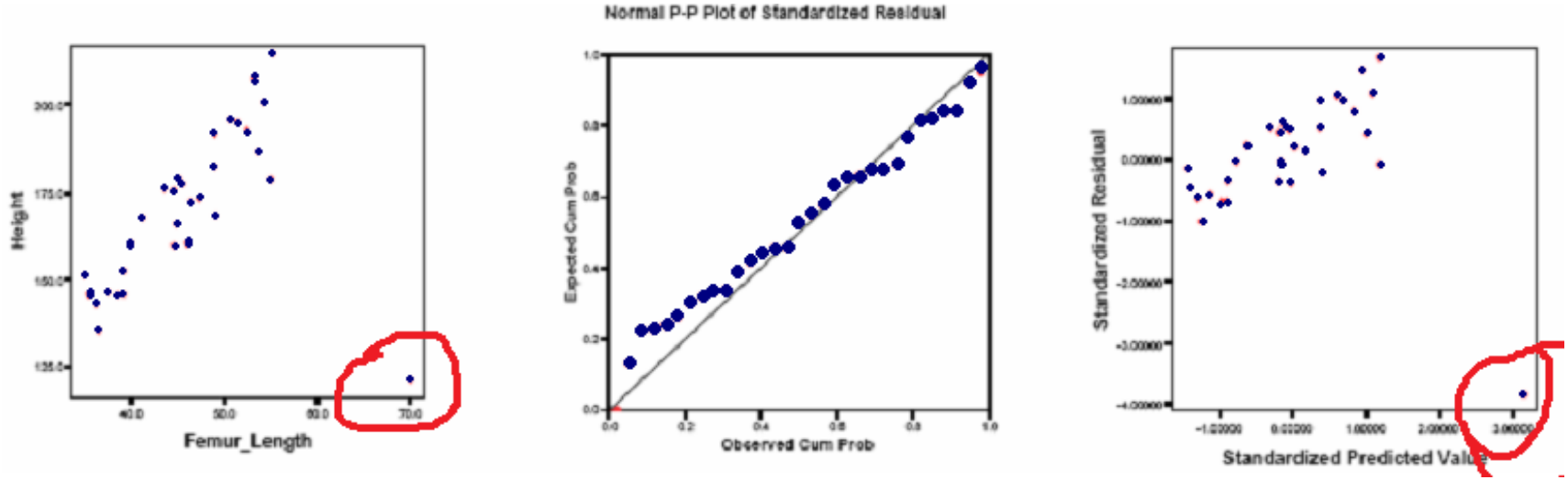
Potential problem # 3

- The variance/standard deviation is not the same for each x value.



Potential problem # 4

- There is an influential point that is obscuring the relationship between x and y.



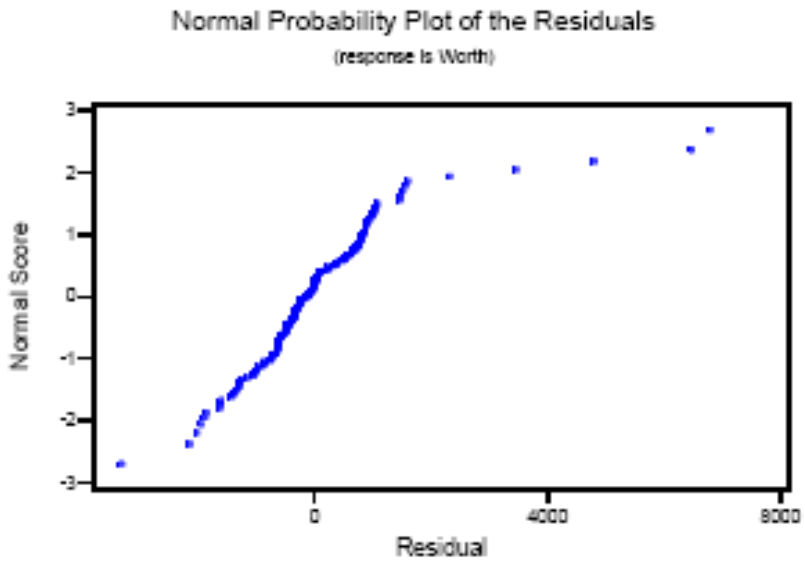
Example: Diamond value v.s. size

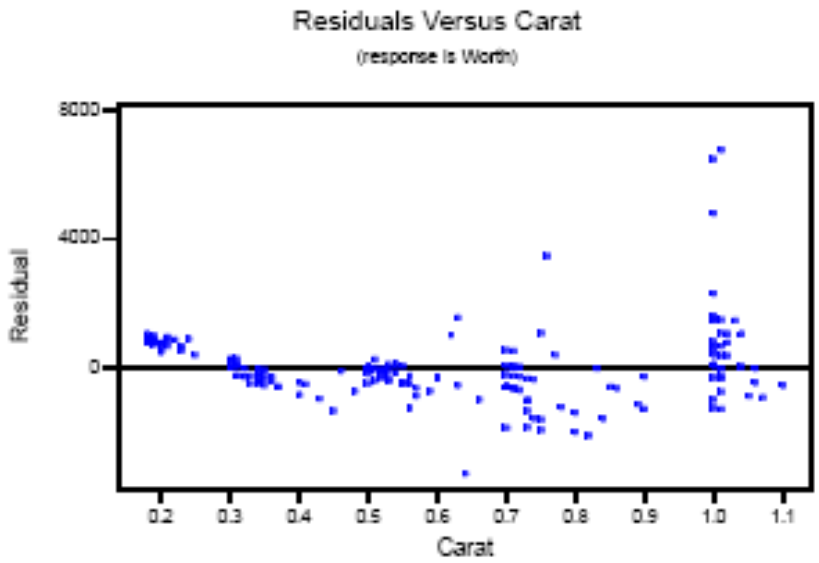
The worth of a diamond depends on several factors, one of which is size, measured in carats. The price should increase as size increases. A simple random sample of 180 diamonds is collected. The diamonds ranged in size from 0.18 carats to 1.07 carats. Before we use the simple linear regression model to study the association between the price of the diamond and its size, we need to do some model checking.

- Explanatory variable: size of a diamond;
- Response variable: the price of the diamond;

Example: Diamond value v.s. size

- Do you see any problems with the model assumptions by examining just the QQ and residual plots?
  - Yes, there is a fan shape in the residual plot;
  - It looks like the variance/standard deviation of y is not the same for each value of x.





- What might be the reason for the problems in the plots?
  - The bigger the diamond, the more variability in the price.
  - Other factors include color, cut and clarity.

Assumptions on the error term ε_i

- For the linear regression model

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \varepsilon_i,\ i = 1, \ldots, n$$

- Conditions on ε_i:
  - Constant variance: $\mathrm{Var}\,\varepsilon_i = \sigma^2$ for all $i = 1, \ldots, n$;
  - ε_i are normally distributed (only assumed for confidence intervals and hypothesis tests)
  - All error terms ε_i's are independent;
- We have learned how to detect the problems, but then how to fix them?

Detection tools for non-constant variance

- The first tool for detecting the non-constant variance is the residual plot

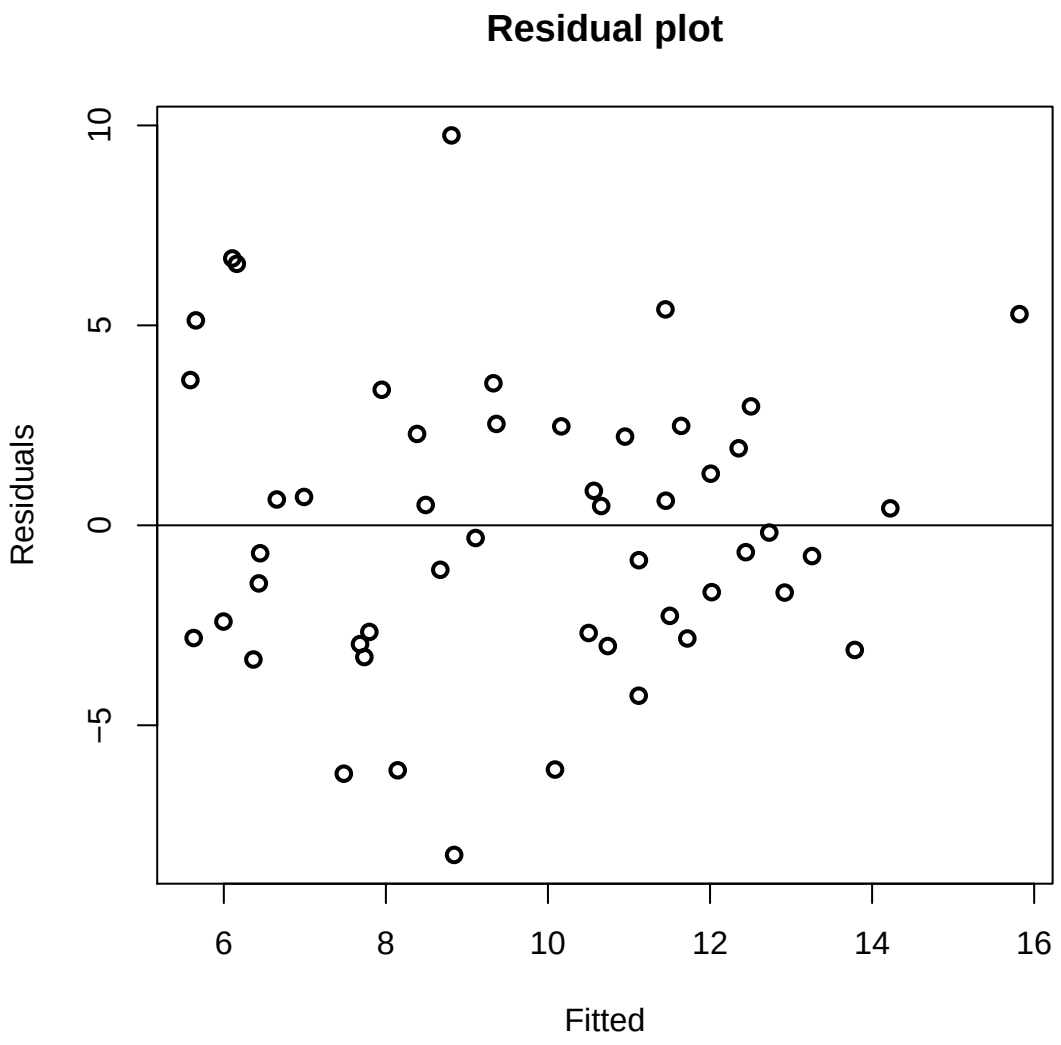
$$\widehat{\varepsilon}_i \text{ (residuals) v.s (the fitted values) } \widehat{y}_i$$

- If no problem, you should observe a random pattern;
- R code for generating residual plots;

```
data(savings,package="faraway")
lmod <- lm(sr~pop15+pop75+dpi+ddpi,savings)
res<-residuals(lmod)#extract residuals
fit<-fitted(lmod)#extract fitted values
#the residual plot
plot(fit,res,xlab="Fitted",ylab="Residuals",main="Residual plot",lwd=2)
abline(h=0) #draw the reference line
```

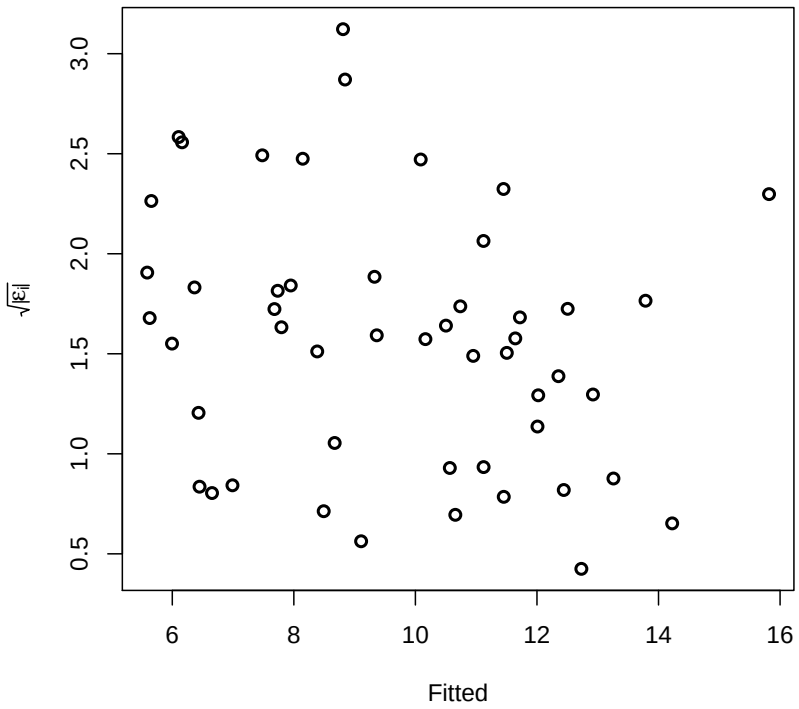
Residual plots for “savings" data

- Does not observe anything suspicious!
- Just to double check, we can use the plot using $\sqrt{|\widehat{\varepsilon}_i|}$ v.s. $\widehat{y}_i$ (more actuate);



Detecting non-constant variance: plot $\sqrt{|\widehat{\varepsilon}_i|}$ v.s. $\widehat{y}_i$

```
plot(fit,sqrt(abs(res)),xlab="Fitted",
ylab=expression(sqrt(abs(epsilon[i]))),lwd=2)
abline(h=0) #draw the reference line
```



- Again, we does not observe anything suspicious!
- The constant variance assumption is probably ok for this model.

Why would there be non-constant variance?

- Assume that the “true" model is $y_i = \alpha + \beta x_i + \gamma z_i + u_i$;
- But we did not observe z_i when we run our regression;
- z_i will get absorbed by the error term and we will actually estimate

$$y_i = \alpha + \beta x_i + \varepsilon_i, \text{where } \varepsilon_i = \gamma z_i + u_i;$$

- If the correlation of x and z is not 0 and$z$ separately affects y (meaning $\gamma \neq 0$), then x is correlated with the error term ε.
- In this case, $\mathrm{Var}(\varepsilon_i)$ may change as x_i changes.

Remedies for non-constant variance of ε

There are couple of remedies non-constant variance for of ε

- Transformation of y:
  - This maybe the most sensible thing to do in practice;
  - Commonly used transformations:

$$\sqrt{y}, \quad \log(y), \quad \log(y+c) \quad , 1/y$$

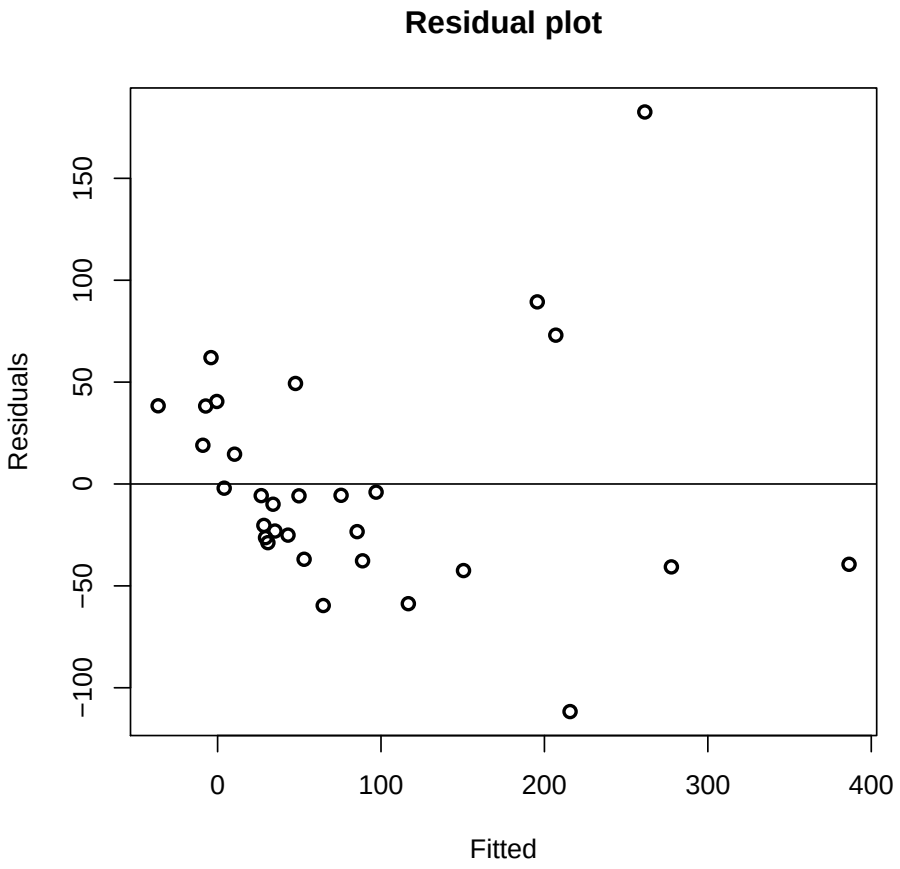
  - How to decide which one to use? Try!
  - Try all commonly used transformations and see which one solves your problem!
- Weighted least square estimator: we will talk about this later;

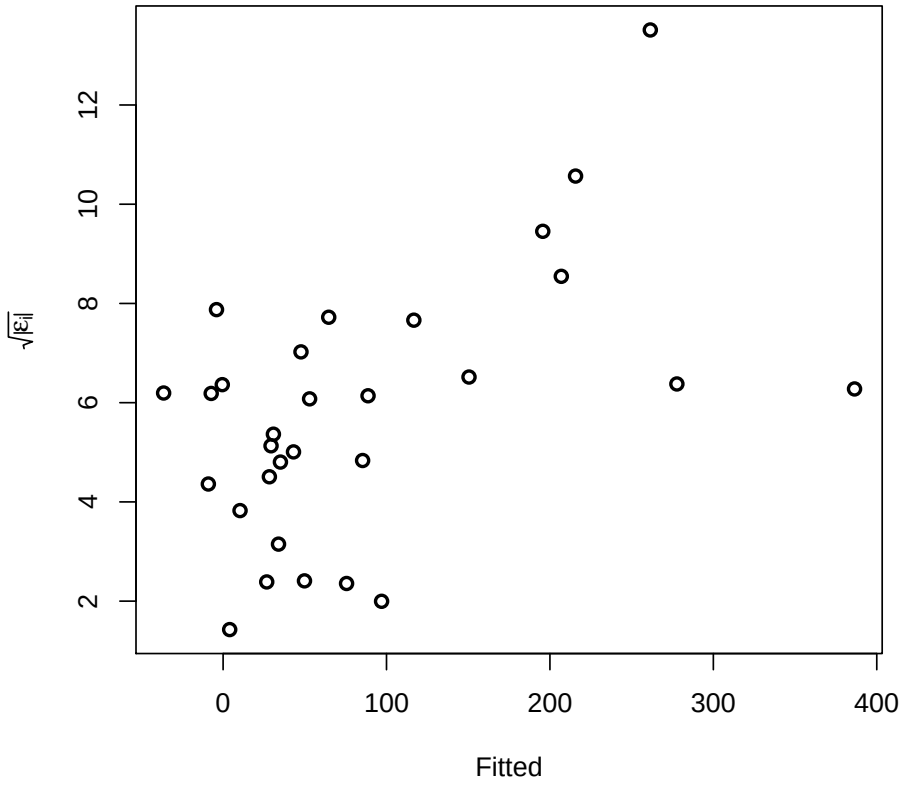
Residual plots for “gala" data

R codes

```
data(gala,package="faraway")
lmod<-lm(Species~Area+Elevation+Scruz+Nearest+Adjacent, gala)
res<-residuals(lmod)#extract residuals
fit<-fitted(lmod)#extract fitted values
#the residual plot
plot(fit,res,xlab="Fitted",ylab="Residuals",main="Residual plot",lwd=2)
abline(h=0) #draw the reference line
##square-root residual plot
plot(fit,sqrt(abs(res)),xlab="Fitted",
ylab=expression(sqrt(abs(epsilon[i]))),lwd=2)
abline(h=0) #draw the reference line
```

Residual plots for “gala" data





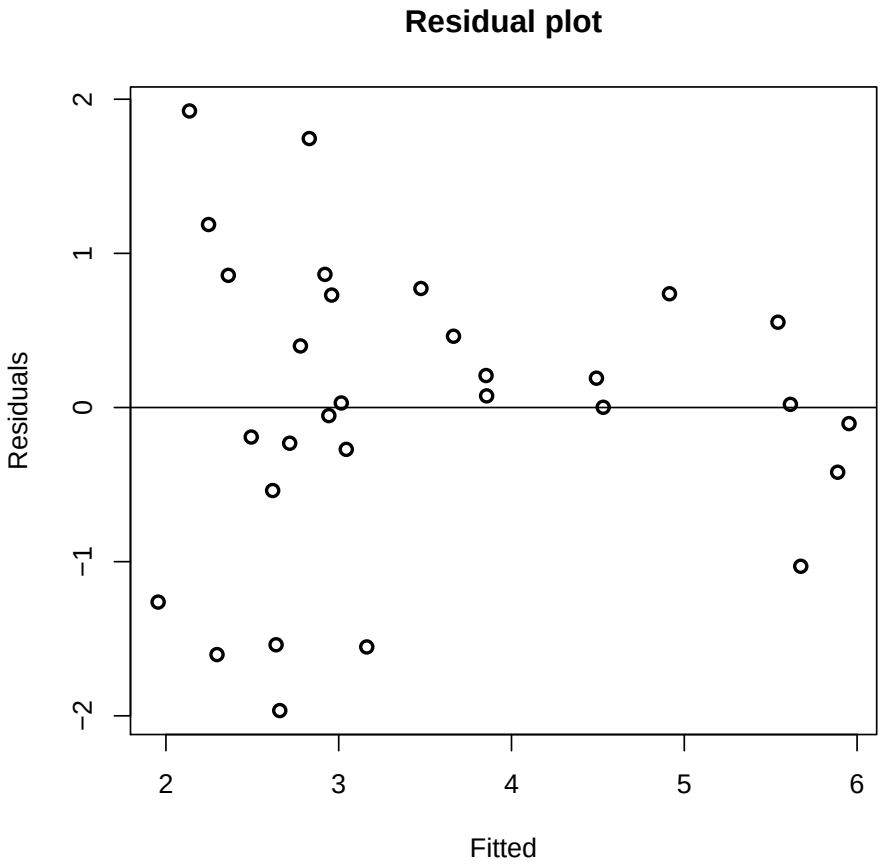
- Both of them show some sort of associations!
- Try transformations!

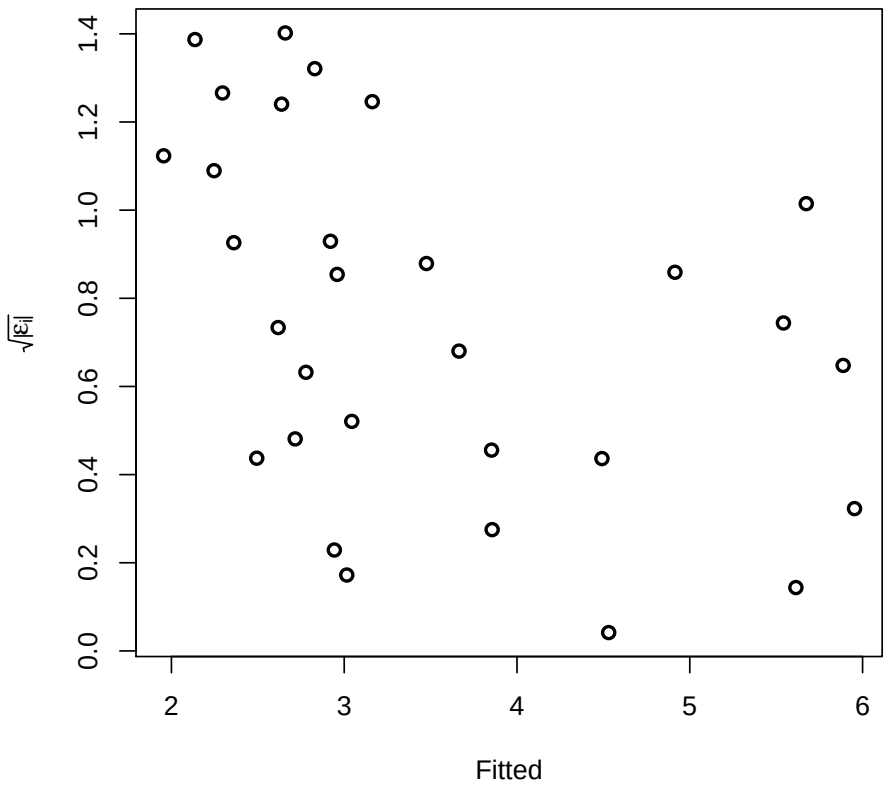
Residual plots after a log transformation

R codes

```
data(gala,package="faraway")
lmod<-lm(log(Species)~Area+Elevation+Scruz+Nearest+Adjacent, gala)
res<-residuals(lmod)#extract residuals
fit<-fitted(lmod)#extract fitted values
#the residual plot
plot(fit,res,xlab="Fitted",ylab="Residuals",main="Residual plot",lwd=2)
abline(h=0) #draw the reference line
##square-root residual plot
plot(fit,sqrt(abs(res)),xlab="Fitted",
ylab=expression(sqrt(abs(epsilon[i]))),lwd=2)
```

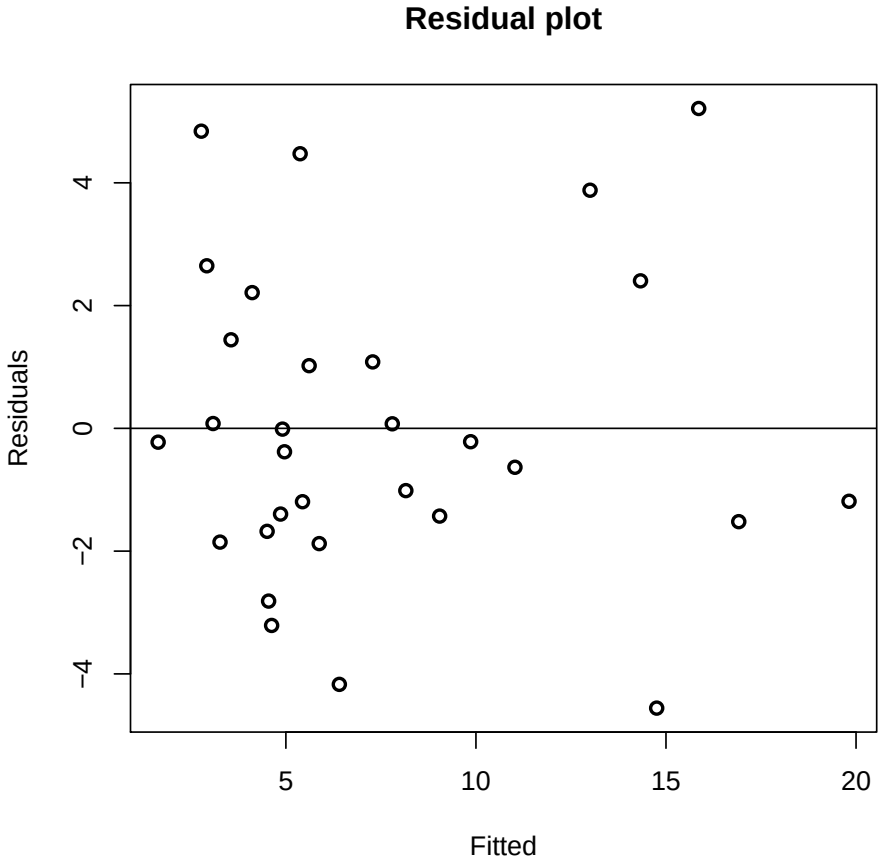
Residual plots after a log transformation

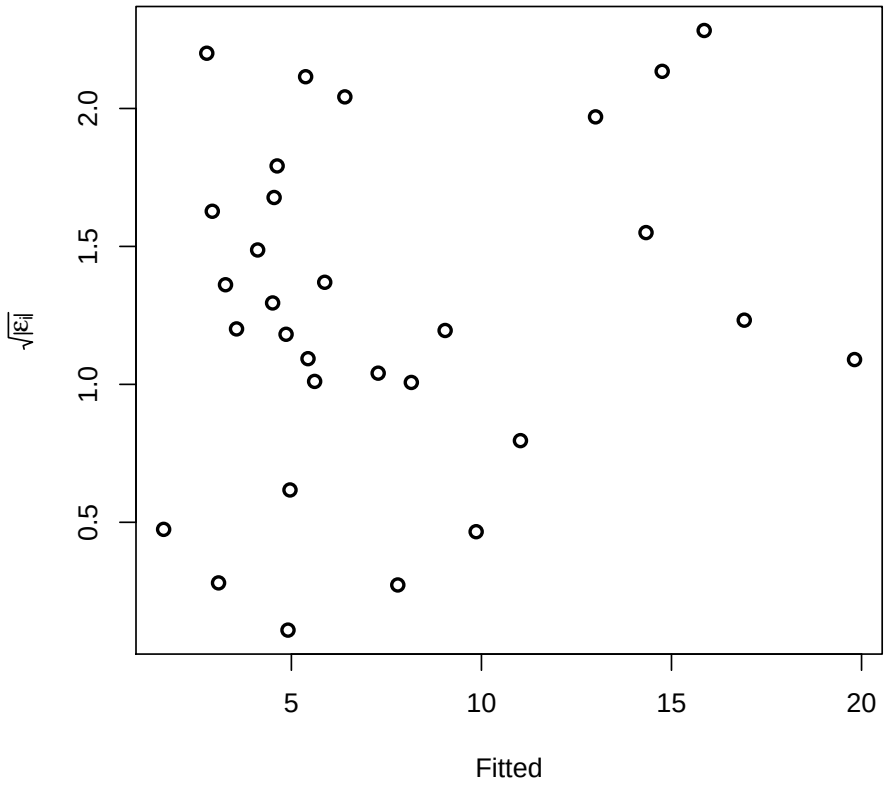




- It is somewhat better than before, but still not quite right!
- Try another transformation: $\sqrt{y}$!

Residual plots after a square-root transformation





- Looks almost random! $\sqrt{y}$ is our guy!
- But you may not get luck every time.
- But even you cannot completely remove the non-constant variance, it does not hurt to make it less severer by using some transformation.

The second problem with ε: the normality

- The normality is only required for confidence intervals and hypothesis testing;
- How to check for normality? Check the q-q plot for residuals $\widehat{\varepsilon}_i$'s.
  - If the qq-plot displays a straight line, then the normality is OK.
  - Otherwise, we need to do something about it;

Normality test: Shapiro-Wilk test

- Shapiro-Wilk test can be used to test the “normality" of the “residuals";
- The R-commend is: shapiro.test();
- The null hypothesis: the data is normally distributed;
- A small p-value means rejection of the normality;
- Shapiro-Wilk is often too sensitive! If your data size n is large, it would reject very small departure from normality;
- Recommendations: We can only recommend this in conjunction with a Q-Q plot at best;
- Q-Q plot is still the main thing you should rely on: remember,
  - Small departure from normal is often OK since we have central limit theorem backing us up;
  - Significant departure from normality is the thing we have to worry about;

QQ plot for “savings" data

The R code for generating qq plots;

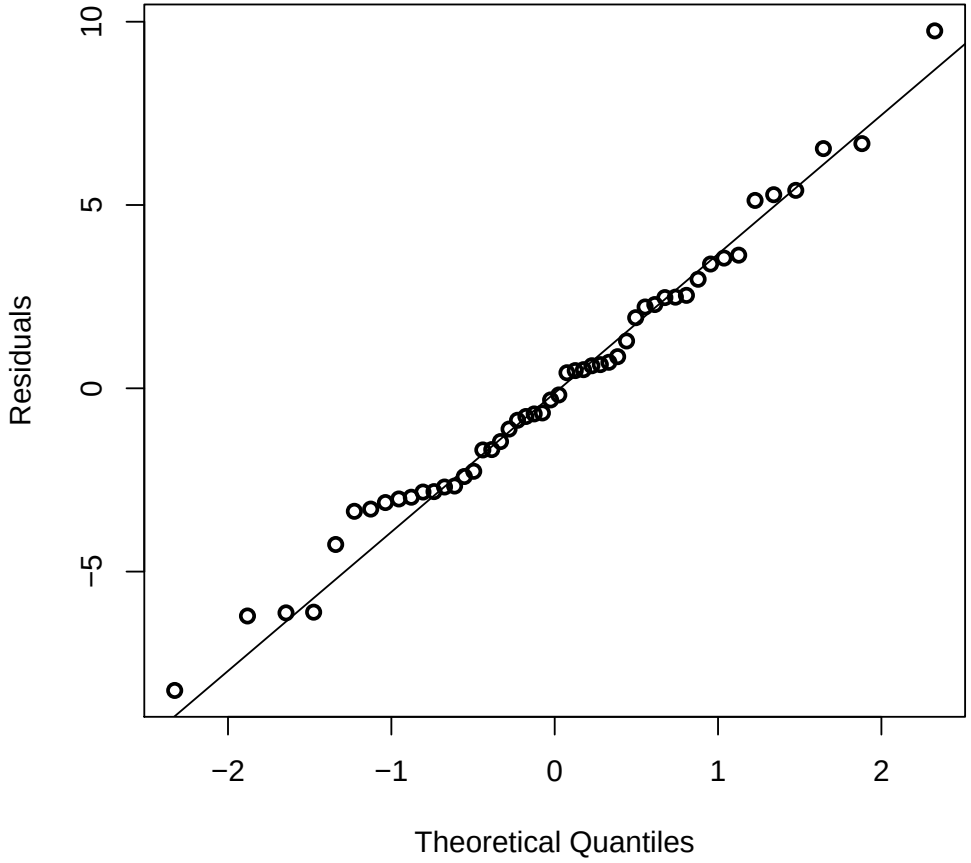
```
data(savings,package="faraway")
lmod <- lm(sr~pop15+pop75+dpi+ddpi,savings)
res<-residuals(lmod)#extract residuals
#the qq plot
qqnorm(res,ylab="Residuals",main="",lwd=2)
qqline(res) #draw a reference line

##Do Shapiro-Wilk test
shapiro.test(res)
> shapiro.test(res)

Shapiro-Wilk normality test

data:  res
W = 0.98698, p-value = 0.8524
```

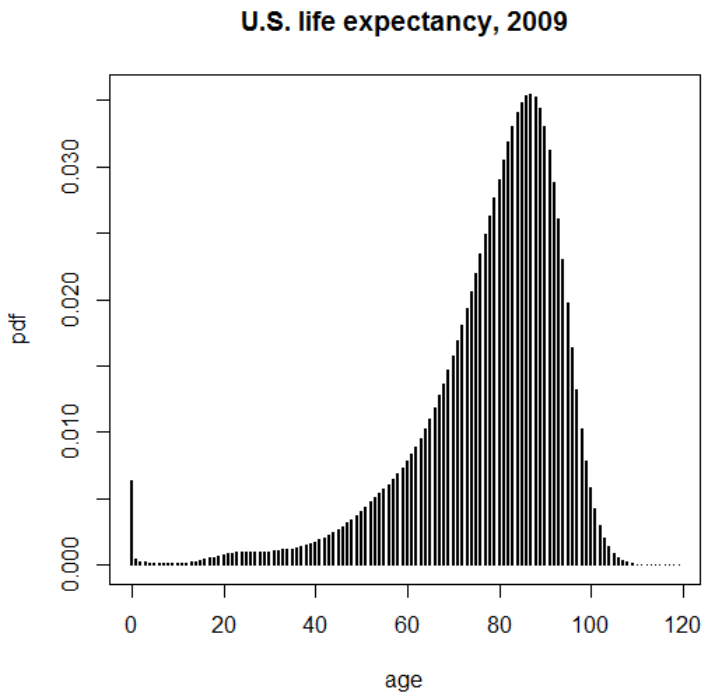
QQ plot for “savings" data

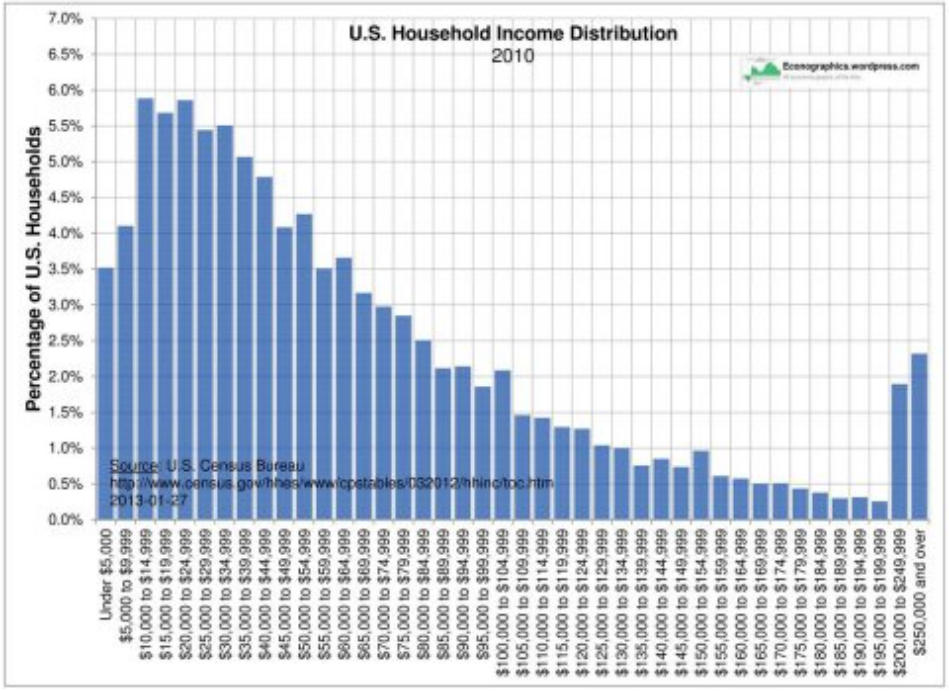


- The QQ-plot is pretty much a straight line;
- Normality seems to be ok;
- In addition, the Shapiro-Wilk test gives a p-value=0.8524, which indicates normality is confirmed;

Some non-normal distributions: skewed distributions

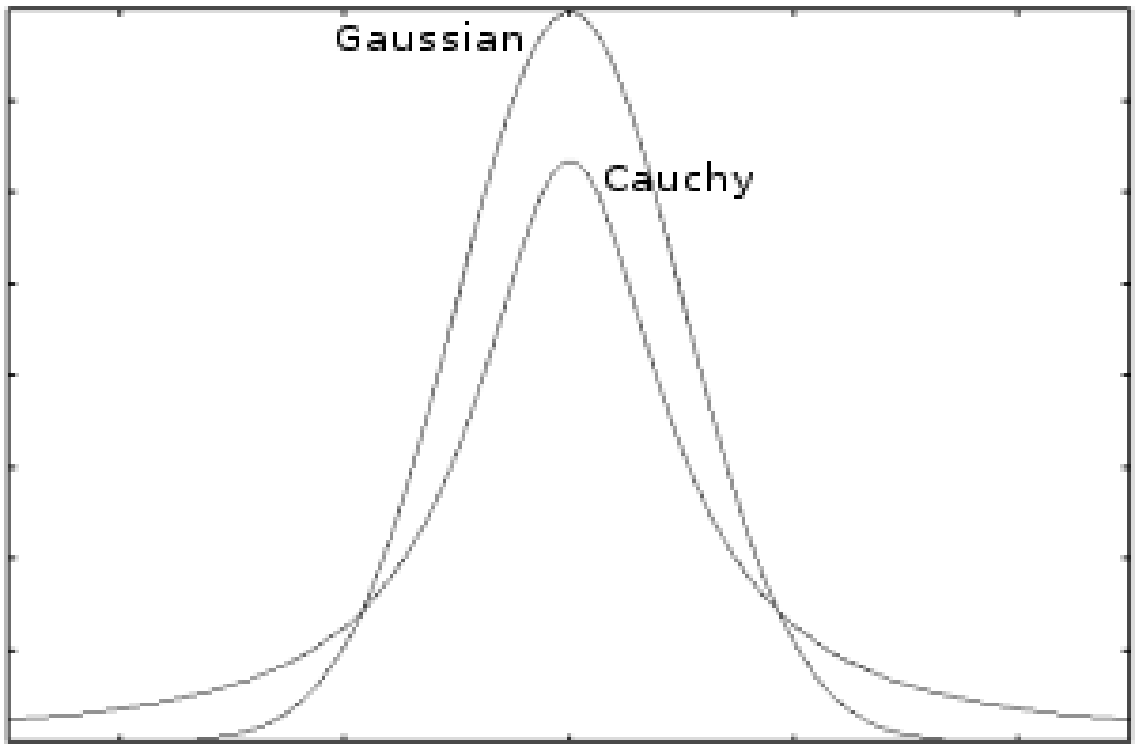
- Left-skewed distribution (life-expectancy): some extreme small values.
- Right-skewed distribution (family income): some extreme large values.





Some non-normal distributions: symmetric heavy-tailed distributions

- More likely to extremely small or large outliers;
- Examples include: student-t distribution, cauchy distribution;
- Really examples: innovation process for stock prices;



Some non-normal distributions: symmetric short-tailed distributions

- Has a lighter tail than the normal distribution;
- Less commonly seen;
- See following qq-plots

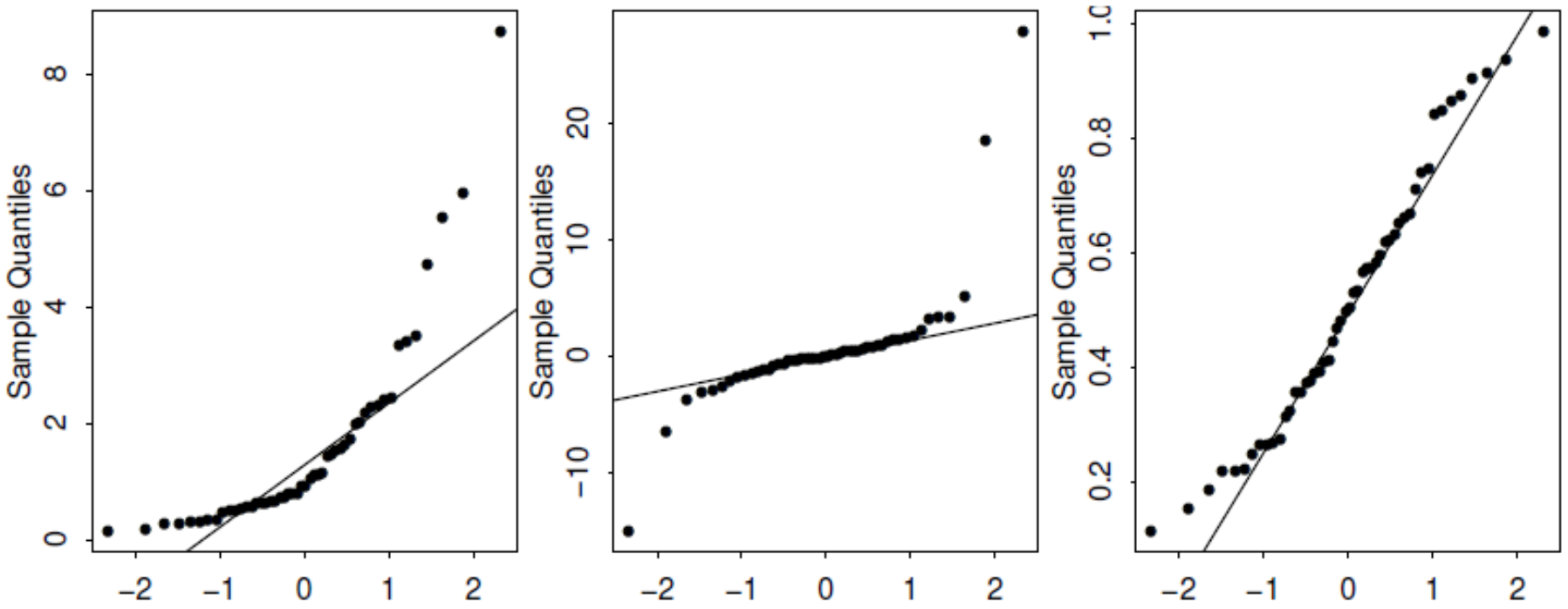


Figure 6.6 Q–Q plots of simulated data for, respectively, a skewed distribution, a long-tailed distribution and a short-tailed distribution.

Recommendations when ε is not normally distributed

The recommendations are case by case

- For a skewed distribution: we can apply a transformation to the response y to make it more normal;
  - One of the most useful tool is the box-cox transformation;

$$y_i^{(\lambda)} = \begin{cases} \dfrac{y_i^{\lambda} - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \ln(y_i) & \text{if } \lambda = 0, \end{cases}$$

  - The parameter λ needs to be picked by other methods;
- For a heavy-tailed distribution:
  - Try to remove some outliers to see if that solves the problem;
  - More robust estimator than the least square estimator should be used;
  - For example, we can use LAD, Huber's loss function, or the quantile regression estimator,etc.
- For short-tailed distribution: do nothing. Won't have big impacts;

A third problem: ε_i's are not independent

- This can only be checked when your data is naturally ordered in time or space:
- Time series data:
- Spatial data:
- You only need to worry about this issue if your response variable have such features;

Example 1: the response y is time series data

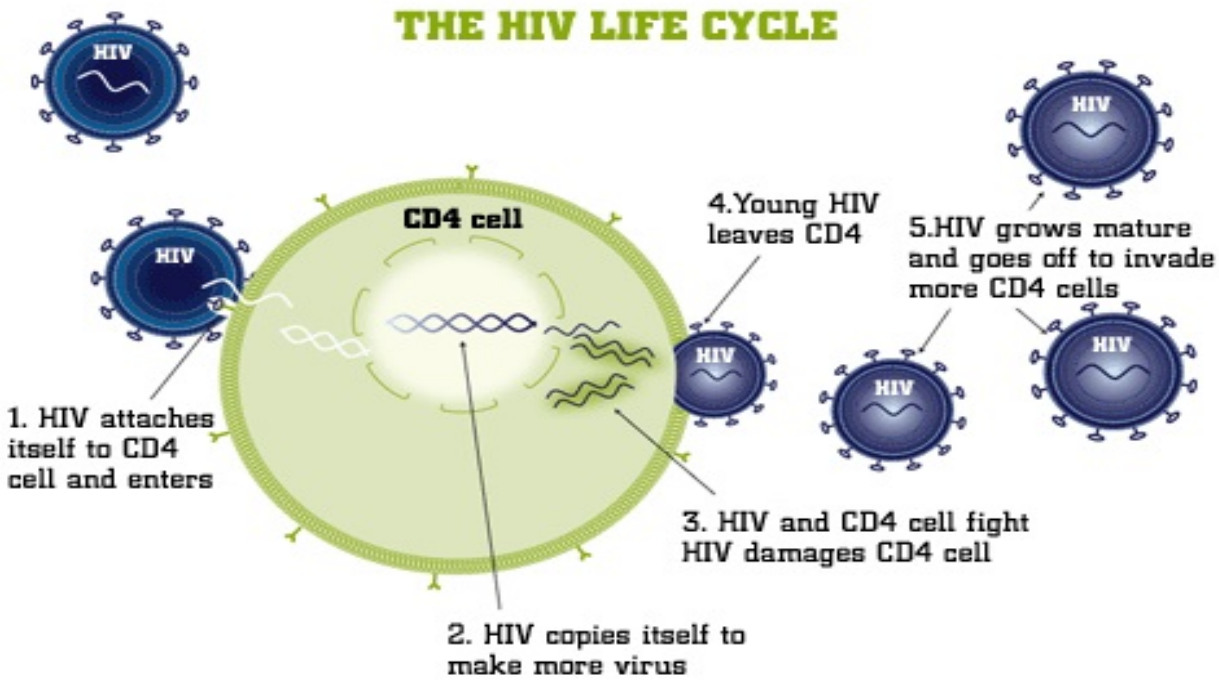
- Many times in practice you will run into data that are recorded in time
  - Daily stock price (y) v.s. other economic factors (GDP, Population, Season, weather, etc..)
  - Daily temperature data v.s. other factors such as (elevation, precipitation,season, etc.)
  - Daily precipitation data v.s. other factors such as (elevation, temperature,season, etc.)
- If a linear regression model is used to model such responses (recorded in time)

$$y_t = \boldsymbol{x}_t^T \boldsymbol{\beta} + \varepsilon_t, \quad t = 1, 2, \ldots, n.$$

- We have a reason to suspect that $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are correlated.

Example 2: the response y is Longitudinal data

- Longitudinal data is a special type of data that is common in many areas. For example
  - CD4 cell: a vital component of the immune system;
  - HIV destroys CD4 cells;
  - CD4 cell level marks the progression of the disease.



Example 2: the response y is Longitudinal data

- As a subset from the Multicenter AIDS Cohort Study;
- Subjects: 283 homosexual men who became HIV-positive between 1984 and 1991;
- Measurements: semi-annual visits (randomly skipped);
  - Response: CD4 percentage measured over time,
  - Covariates of interest:

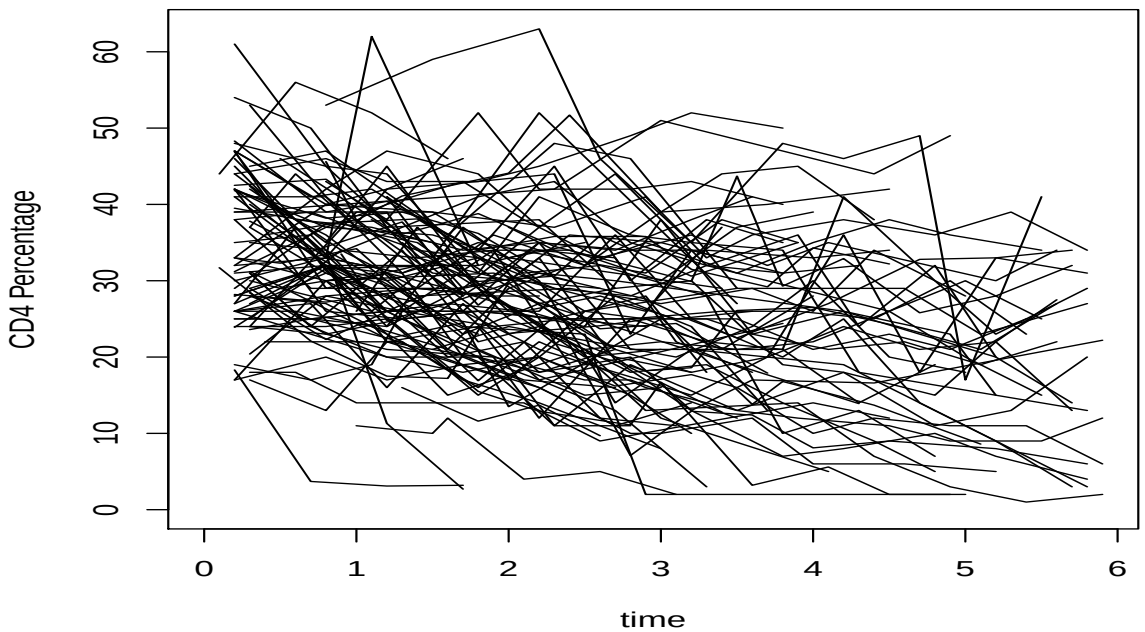
    Age: age at HIV infection;
    Smoke: smoker (1) and nonsmoker (0);
    Pre-infection CD4: CD4 percentage at the time of HIV infection.
- Question: how do these covariates contribute to the progress of the disease?

Example 2: the response y is Longitudinal data

Longitudinal (Cluster) data $(y_{ij}, \boldsymbol{X}_{ij})$, $j = 1, \cdots, n_i$, $i = 1, \cdots, n$.

- Within subject observations are correlated, denote $\mathbf{y}_i = (y_{i1}, \cdots, y_{in_i})^T$ and $\varepsilon_i = (\varepsilon_{i1}, \cdots, \varepsilon_{in_i})^T$;
- $\mathbf{y}_1, \cdots, \mathbf{y}_n$ are independent;
- A regression model can be used as

$$y_{ij} = \boldsymbol{x}_{ij}\boldsymbol{\beta} + \varepsilon_{ij},$$
$$cov(\boldsymbol{\varepsilon}_i) = \boldsymbol{\Sigma}_i.$$

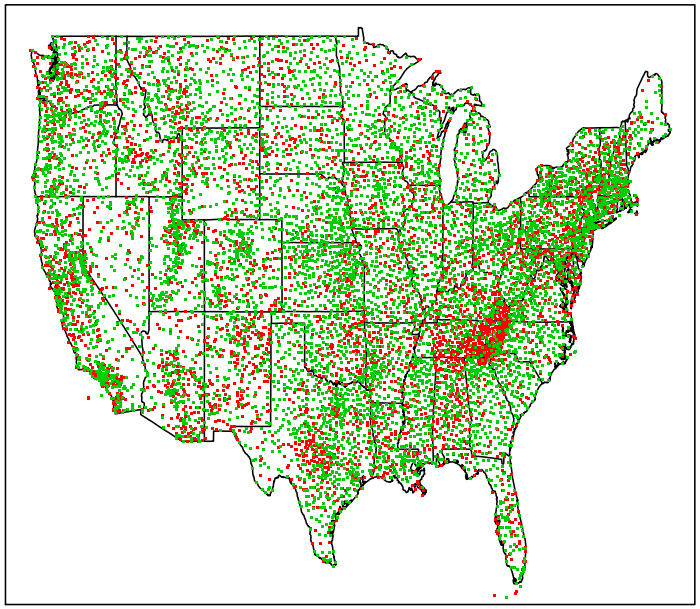


Example 3: the response y is Spatial data

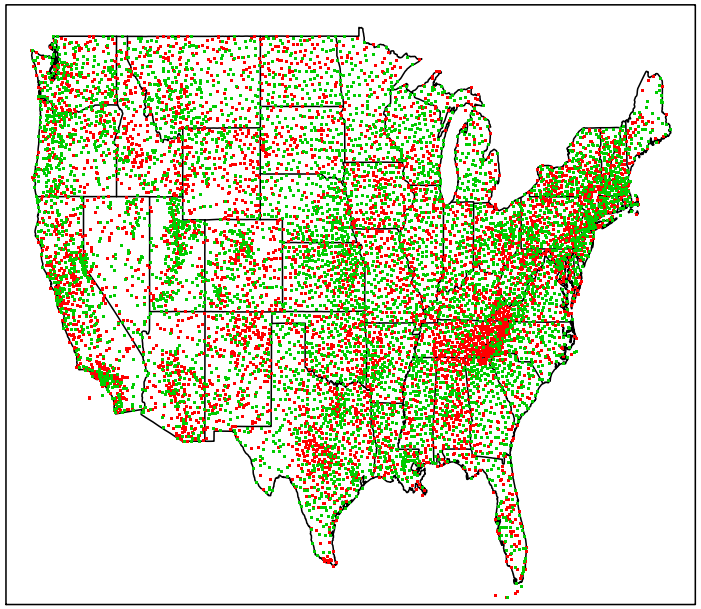
Annual total precipitation data (y) for the United States.

- 11,918 stations throughout the United States.
- From 1895–1997: 103 years.
- About 6000–7000 stations with observations each year.

(a) Station locations (1980), 6905



(b) Station locations (1981), 5744



Example 3: the response y is Spatial data

We can model the relationship between the precipitation y and other factors

$$Y(\boldsymbol{s}) = \beta_0 + \beta_1 x_1(\boldsymbol{s}) + \cdots + \beta_p x_p(\boldsymbol{s}) + \varepsilon(\boldsymbol{s}).$$

- For precipitation: $x_j(\boldsymbol{s})$'s can be elevation, local temperature, longitude, latitude, coast (Florida) V.S. inland (Nevada), etc.
- $\varepsilon(\boldsymbol{s})$: spatially correlated Gaussian process with mean 0;
- In such cases, $\varepsilon(\boldsymbol{s})$'s are correlated!

Strategies with correlated error terms ε_i's

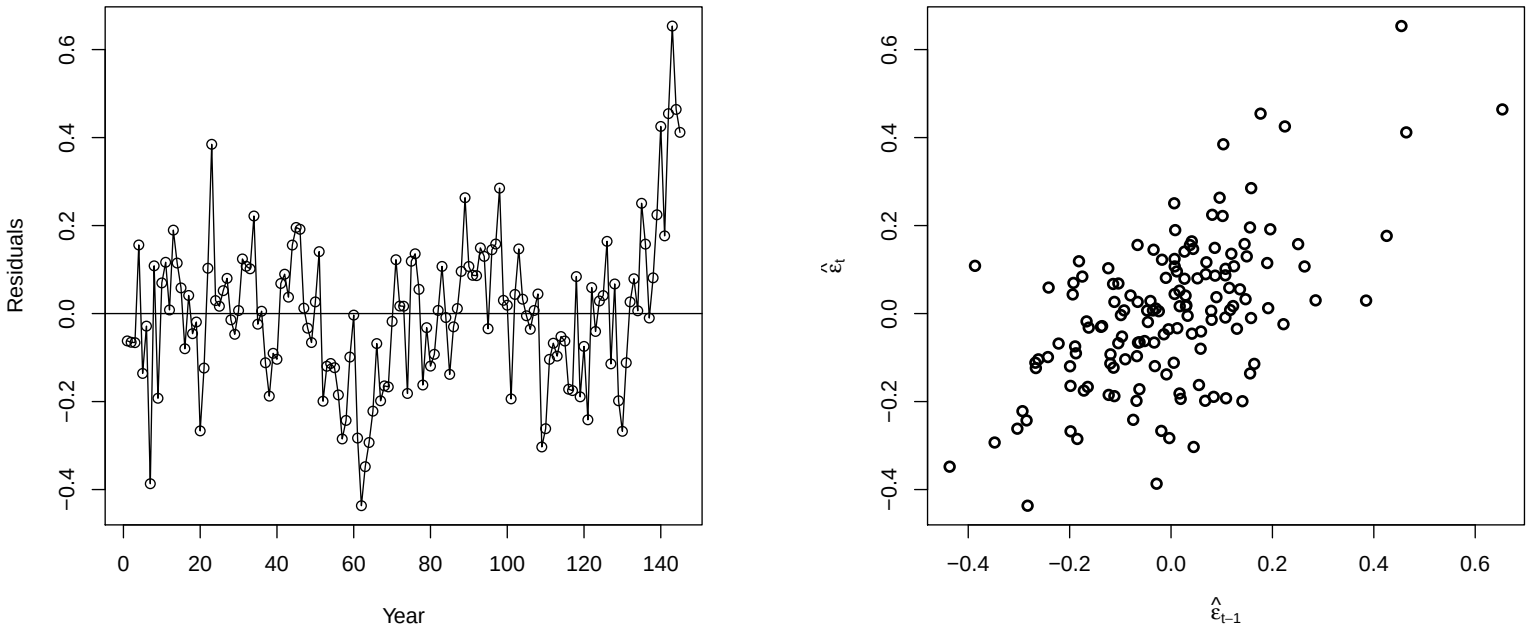
- There is NO general way to tell whether ε_i's are correlated or not since they are not observable.
- However, when you have following three types of data in your hand, it is common sense to suspect correlation among ε_i's;
  - Time series response;
  - Longitudinal data;
  - Spatial data (or spatio-temporal data);
- But how to check? It is really case by case.
- For time series data, you can have two graphical tools
  - The plot with residuals v.s. time;
  - Plot successive pair of residuals;

Example: “globwarm" data

R codes

```
data(globwarm,package="faraway")
lmod <- lm(nhtemp~wusa+jasper+westgreen+chesapeake+
tornetrask+urals+mongolia+tasman, globwarm)
res<-residuals(lmod)#extract residuals
#the residual plot
plot(res,type="l", xlab="Year",ylab="Residuals")
points(res)
abline(h=0) #draw the reference line
##scatter plot of successive pair of residuals
plot(res[-length(res)],res[-1],xlab=expression(hat(epsilon)[t-1]),
ylab=expression(hat(epsilon)[t]),lwd=2)
```

Example: “globwarm" data



Couple of observations

- The left panel: a sequence of consecutive positive numbers followed by another sequence of consecutive negative numbers; This is an indication of potential series correlation;
- Right panel: there is an indication of strong correlation between $\hat{\varepsilon}_t$ and $\hat{\varepsilon}_{t-1}$, which confirms the findings in the residual scatter plot.–This type of correlation is usually called autocorrelation.

Test for autocorrelation: Durbin-Watson test

- The test statistics is defined as

$$DW = \frac{\sum_{i=2}^{n}(\hat{\varepsilon}_i - \hat{\varepsilon}_{i-1})^2}{\sum_{i=1}^{n}\hat{\varepsilon}_i^2}$$

- It is closely related to the correlation coefficient between $\hat{\varepsilon}_i$ and $\hat{\varepsilon}_{i-1}$;
- The null hypothesis is: there is no autocorrelation;
- A rejection of this test: means there is potentially some sort of auto correlation.
- The R code for the DW test: dwtest();

R codes

```
library(lmtest) ##You need to load a R package to use this command
data(globwarm,package="faraway")
dwtest(nhtemp ~ wusa+jasper+westgreen+chesapeake+tornetrask
+urals+mongolia+tasman, data=globwarm)

Durbin-Watson test
data:  nhtemp ~ wusa+jasper+westgreen+chesapeake+tornetrask+urals+mongolia+
DW = 0.81661, p-value = 1.402e-15
alternative hypothesis: true autocorrelation is greater than 0
```

Recommendations when ε_i's are correlated

The recommendations are case by case

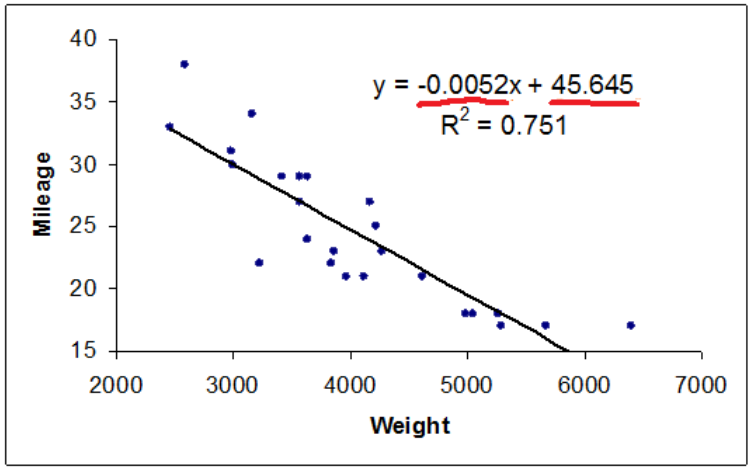
- If you can identify the root cause of correlation, remove the cause of the correlation;
  - Sometimes, serial correlation can be caused by a missing covariate;
  - Assume that the "true" model is $y_i = \alpha + \beta x_i + \gamma z_i + u_i$;
  - But we omit z_i (perhaps because we don't have a measure for it) when we run our regression;
  - z_i will get absorbed by the error term and we will actually estimate

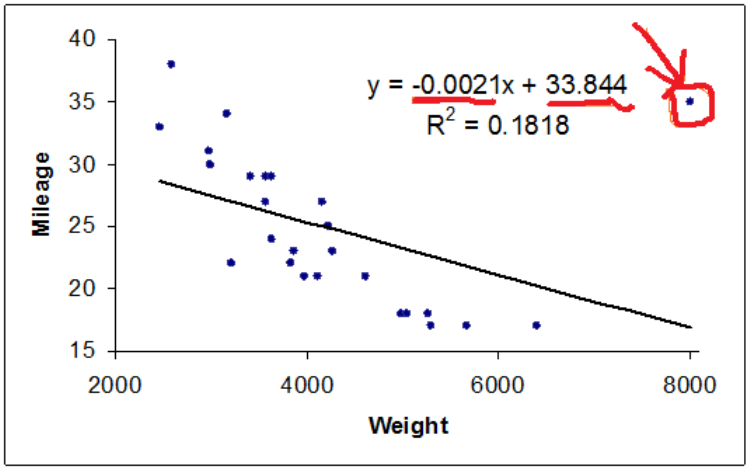
$$y_i = \alpha + \beta x_i + \varepsilon_i, \text{where } \varepsilon_i = \gamma z_i + u_i;$$

  - ε_i may be correlated because of the existence of z_i;
  - Put z_i back to the mean structure can help ε_i's become independent;
  - But this is usually very hard to do!
- If you cannot identify the source of correlation, then build the correlation into your estimation!
  - If your data are highly correlated, you essentially have less amount new information;
  - It is equivalent to having a smaller sample size;
  - With a smaller sample size, the variance of $\widehat{\boldsymbol{\beta}}_{ols}$ is larger;
  - We can the variance smaller by using Generalized least square estimator.

Outliers and influential points

- If there is only one predictor in the regression model, identifying the influential points are straightforward;
  - By looking at a scatter plot of y v.s. x;





- The **high leverage points** are those with extreme x values;
- But when we have multiple predictors, how to define influential points/outliers?

Leverages

- The leverages h_i's are diagonal elements of $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$;
- Recall that $\widehat{\boldsymbol{Y}} = \mathbf{H}\boldsymbol{Y}$, consider the ith observation, we have

$$\hat{y}_i = H_{i1}y_1 + H_{i2}y_2 + \cdots + H_{ii}y_i + \cdots + H_{in}y_n$$

- The leverage $h_i = H_{ii}$ is the amount of weight of the predicted value $\hat{y}_i$ put on y_i;
- The leverage h_i is a measure of self-influence of y_i on $\hat{y}_i$; one can show that

$$cov(\hat{y}_i, y_i) = H_{ii}\sigma^2 = h_i\sigma^2.$$

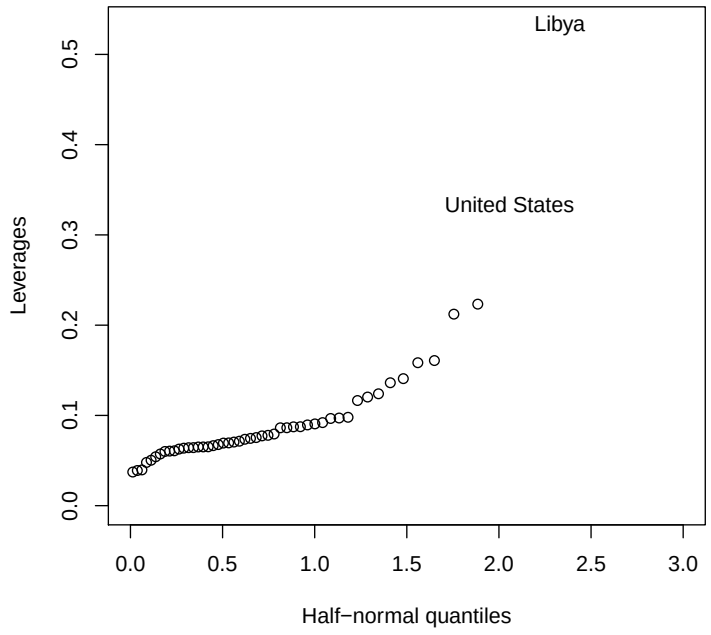
- One can mathematically prove that $h_1 + \cdots + h_n = p + 1$;
- A rough rule is that observations with a leverage of more than $2(p+1)/n$ should be looked at more closely. That is, they are influential points; but why?

Graphical representation of the leverages

- We have side that observations with a leverage of more than $2(p+1)/n$ should be looked at more closely.
- The R function for extracting leverages: *hatvalues()*;
- We can also use a graphical tool to pick out abnormal leverages: Half-normal plot;
- Half-normal plot is similar to a Q-Q plot, except that it uses quantiles of half-normal distribution as x-axis;
- The R function for generating leverage-half-normal plot: *halfnorm()* in package Faraway;

```
library(faraway)
data(savings,package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
lev <- hatvalues(lmod) #extract the leverage values
countries <- row.names(savings)
halfnorm(lev,labs=countries,ylab="Leverages",xlim=c(0,3))
```

Graphical representation of the leverages



- We are not looking for a linear pattern in the half-normal plot;
- Identify outliers/influential points by picking out abnormal points in this plot!
- In this case, Libya and United States seems to be very influential/abnormal;

Identify influential points: Cook's distance

- An influential point is one whose removal from the dataset would cause a large change in the model fit.
- In other words, to identify influential points, we need to evaluate how much change in model fit can be caused by removing one point
- The Cook's distance is such a measure, defined as

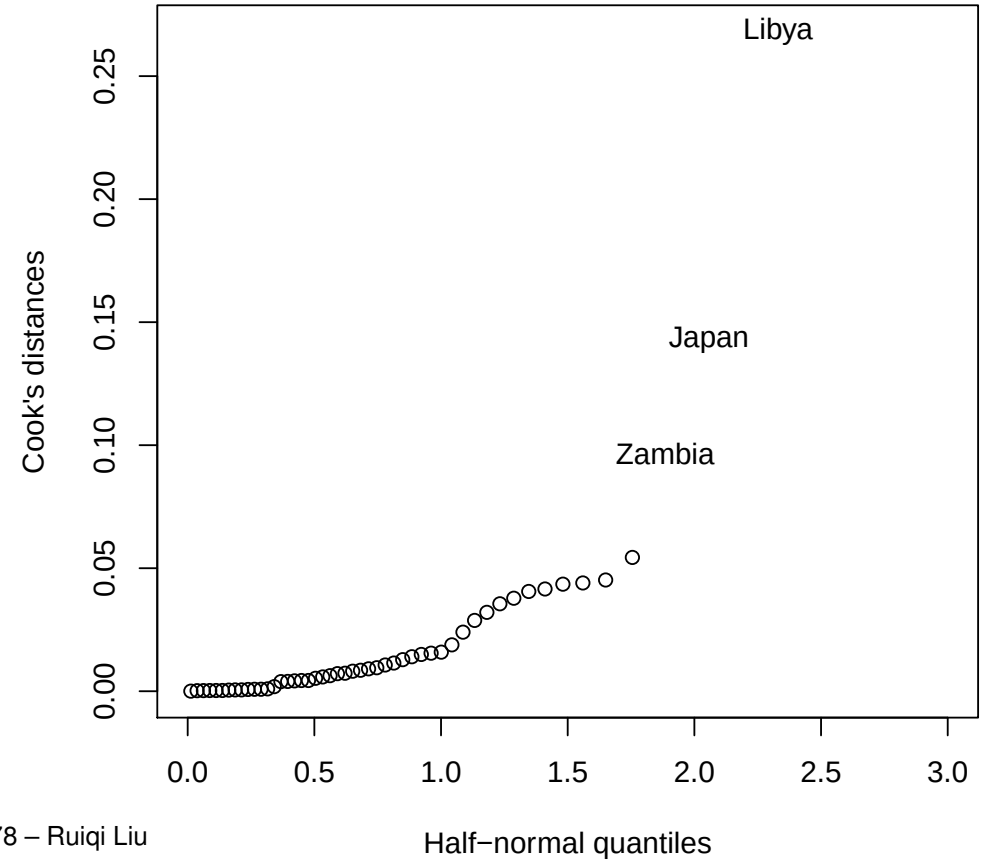
$$D_i = \frac{(\hat{\boldsymbol{Y}} - \hat{\boldsymbol{Y}}_{(i)})^T(\hat{\boldsymbol{Y}} - \hat{\boldsymbol{Y}}_{(i)})}{(p+1)\hat{\sigma}^2} = \frac{\hat{\epsilon}_i^2}{(p+1)\hat{\sigma}^2}\frac{h_i}{(1-h_i)^2},$$

  where p is the number of predictors in the model.
  - $\hat{\boldsymbol{Y}}$: predicted values for the entire data set using all data;
  - $\hat{\boldsymbol{Y}}_{(i)}$: predicted values for the entire data set using all data without observation i;
- Note that the Cook's distance D_i for the ith observation is a function of its leverage h_i.

Using R to calculate Cook's distance

- The R command for Cook's distance: *cooks.distance()*

```
library(faraway)
data(savings,package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
cook <- cooks.distance(lmod)#find Cook's distance
#half-normal plot of Cook's distance
halfnorm(cook,3,labs=countries,ylab="Cook's distances",xlim=c(0,3))
```



Identify outliers: studentized residual

FINDING UNUSUAL OBSERVATIONS 87

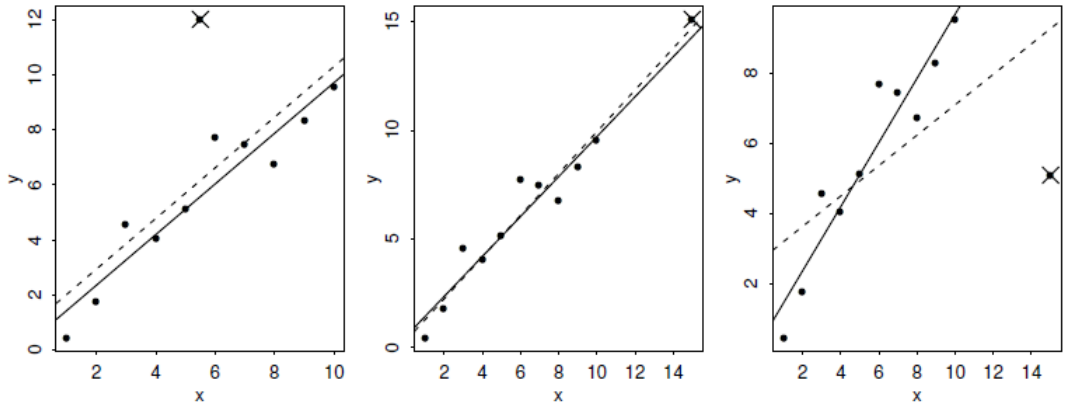


Figure 6.9 In all three plots, the additional point is marked with a cross. The solid line is the fit using the 10 original points and the dashed line is the fit with the added point.

- Let $\widehat{\boldsymbol{\beta}}_{(i)}$ and $s_{(i)}$ be the ols estimator and the estimated error standard deviation of the regression model with ith case deleted;
- Then the prediction of y_i with the ith case deleted is $\hat{y}_{(i)} = x_i^T \widehat{\boldsymbol{\beta}}_{(i)}$;
- For outliers, $\hat{y}_{(i)} - y_i$ is large; (You need be careful here: the underlying assumption is after removing the outlier, all remaining data follow the same model, which might not be true if you have several outliers;especially when they are close to each other)

Identify outliers: studentized residual

- So to test whether a point $(\boldsymbol{x}_i, y_i)$ is an outlier, essentially we need a measure of how large $\hat{y}_{(i)} - y_i$ is;
- Define the statistic (named studentized residual)

$$t_i = \frac{\hat{y}_{(i)} - y_i}{se(\hat{y}_{(i)} - y_i)} = \frac{\hat{y}_{(i)} - y_i}{s_{(i)}\sqrt{1 + \boldsymbol{x}_i^T(\mathbf{X}_{(i)}^T\mathbf{X}_{(i)})^{-1}\boldsymbol{x}_i}}.$$

- Calculation of t_i's are not easy, we need to estimate the regression n times; (each time leave on observation/case out);
- Luckily, we have a short-cut formula for t_i

$$t_i = \frac{\hat{\varepsilon}_i}{s_{(i)}\sqrt{1 - h_i}} = r_i\sqrt{\frac{n - p - 2}{n - p - 1 - r_i^2}}$$

where $r_i = \frac{\hat{\varepsilon}_i}{s\sqrt{1-h_i}}$. (no need to know how to get this.)

Identify outliers: studentized residual

- We can calculate a t_i for each data points;
- R function for that:*rstudent()*;
- For a reasonable sample size n, $|t_i| \geq 3$ will be considered an outlier;
- For $2 \leq |t_i| \leq$, it maybe suspicious depending on α and n;
- Many high $|t_i|$'s indicates inadequate model.

```
library(faraway)
data(savings,package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
studres <- rstudent(lmod)#standardized residuals
studres[which.max(abs(studres))]
Zambia
2.853558
range(studres)
[1] -2.313429  2.853558
```

The studentized residuals may fail with multiple outliers

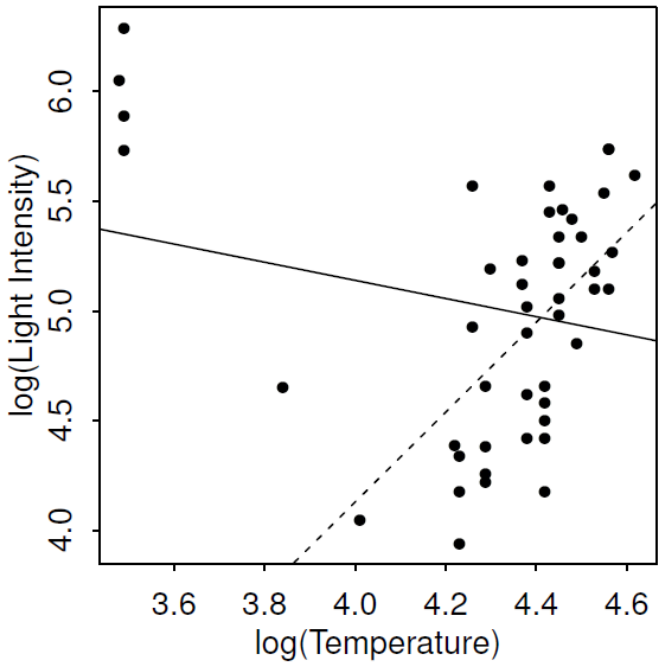


Figure 6.10 Regression line including four leftmost points is solid and excluding these points is dotted.

- The range of studentized residuals is $[-2.0494, 1.9058]$;
- Based on the rule, there is no outliers!
- Studentized residuals has failed to identify outliers!
- Reason: those outliers are grouped together! (you need to keep this in mind when using studentized residuals to identify outliers.)

Diagnostic plots for regression in R

```
library(faraway)
data(savings,package="faraway")
lmod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, savings)
plot(lmod) # produce diagnostic plots
```

Diagnostic plots for regression in R

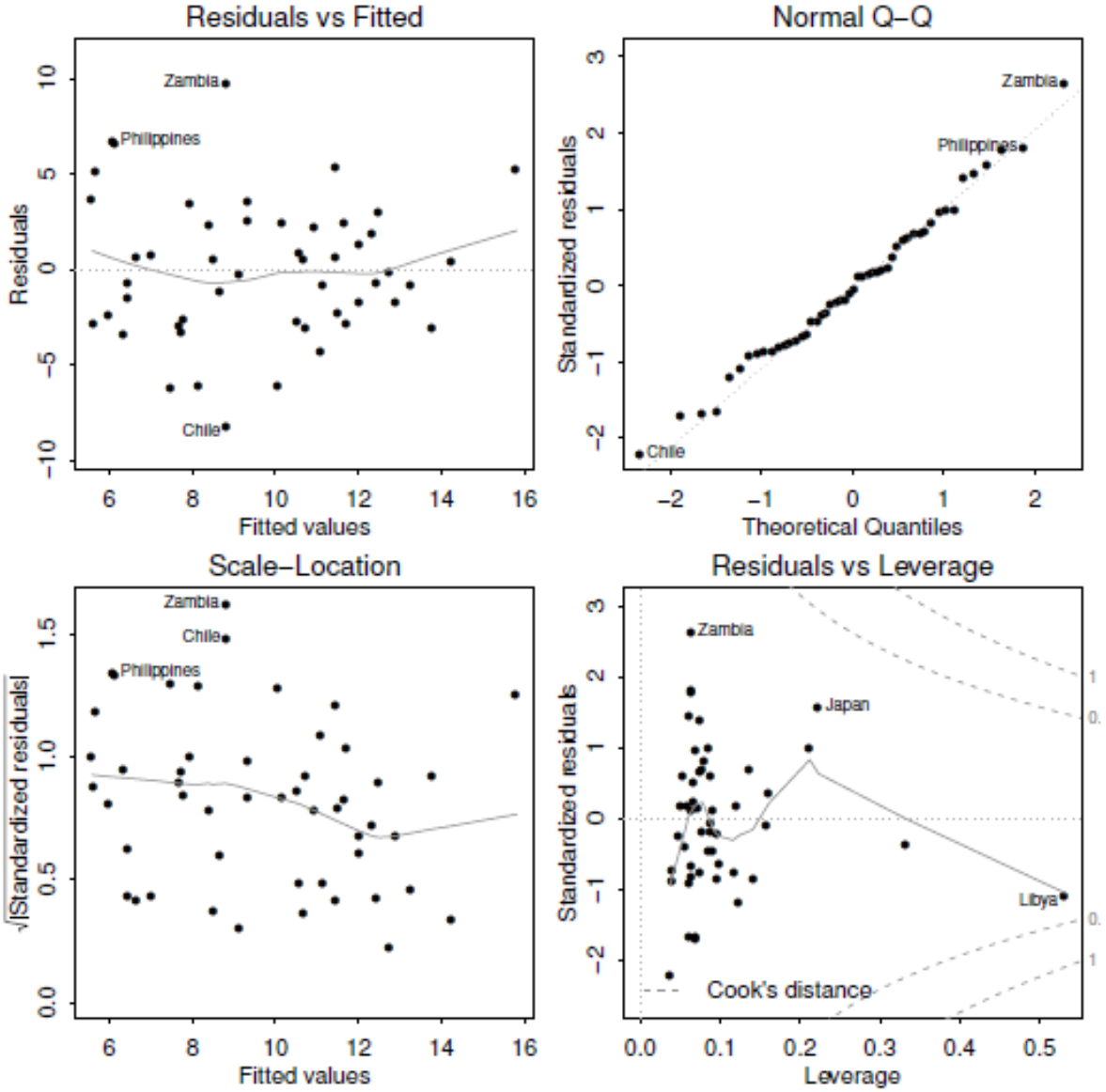


Figure 6.12 Diagnostic plots for the savings data

Pairwise scatter plot

- We would also need to examine the linear relationship $\boldsymbol{Y} = \mathbf{X}^T\boldsymbol{\beta} + \boldsymbol{\varepsilon}$;
- Sometimes, the residual plots can be helpful to look at
  - some types of nonlinear relationship between y and x's may be reflected in the non-linear pattern of the residual plot (i.e. $\hat{\varepsilon}_i$'s v.s. $\hat{y}_i$'s);
- It is also helpful to look at the scatter plots of y against each x_j's;
- However, the linear relationship between y and x_j in a linear regression model is conditional on other predictors; so y and x_j may not be linear;
  - Consider the model
    $$y = \beta_0 + x_1\beta_1 + x_2\beta_2 + \varepsilon$$
    where $x_2 = x_1^2$;
  - Conditional on x_2, y and x_1 is linearly related;
  - However, without x_2, the relationship between y and x_1 is NOT linear!

Partial regression plots

- To check the relationship between y and x_j, we need to first isolate the relationship between them;
- A partial regression plots:
  - First regress y on all x's except x_j, get residuals $\widehat{\boldsymbol{\delta}}$;
  - Then regress x_j on all other x's, get residuals $\widehat{\boldsymbol{\gamma}}$;
  - Plot residuals $\widehat{\boldsymbol{\delta}}$ against residuals $\widehat{\boldsymbol{\gamma}}$;
  - Look for non-linearity/outliers/influential points;

```
library(faraway)
d <- residuals(lm(sr ~ pop75 + dpi + ddpi,savings))
m <- residuals(lm(pop15 ~ pop75 + dpi + ddpi,savings))
plot(m,d,xlab="pop15 residuals",ylab="Savings residuals",pch=16)
lmod <- lm(d~m)
abline(lmod)
```

Partial residual plots

- Partial residual plots (as opposed to the partial regression plots):
  - First regress y on all x's with the fitted model as

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p;$$

  - Compute the partial residual $\hat{\varepsilon}_{partial} = y - \sum_{k \neq j} \hat{\beta}_k x_k$;
  - Plot partial residuals $\hat{\varepsilon}_{partial}$ against x_j;
  - R command: *termplot*
  - Look for non-linearity/outliers/influential points;

```
library(faraway)
lmod <- lm(sr ~ pop15+pop75 + dpi + ddpi,savings)
termplot(lmod,partial.resid=TRUE,terms=1,pch=16)
```

Partial Regression Plot V.S. Partial Residual Plot

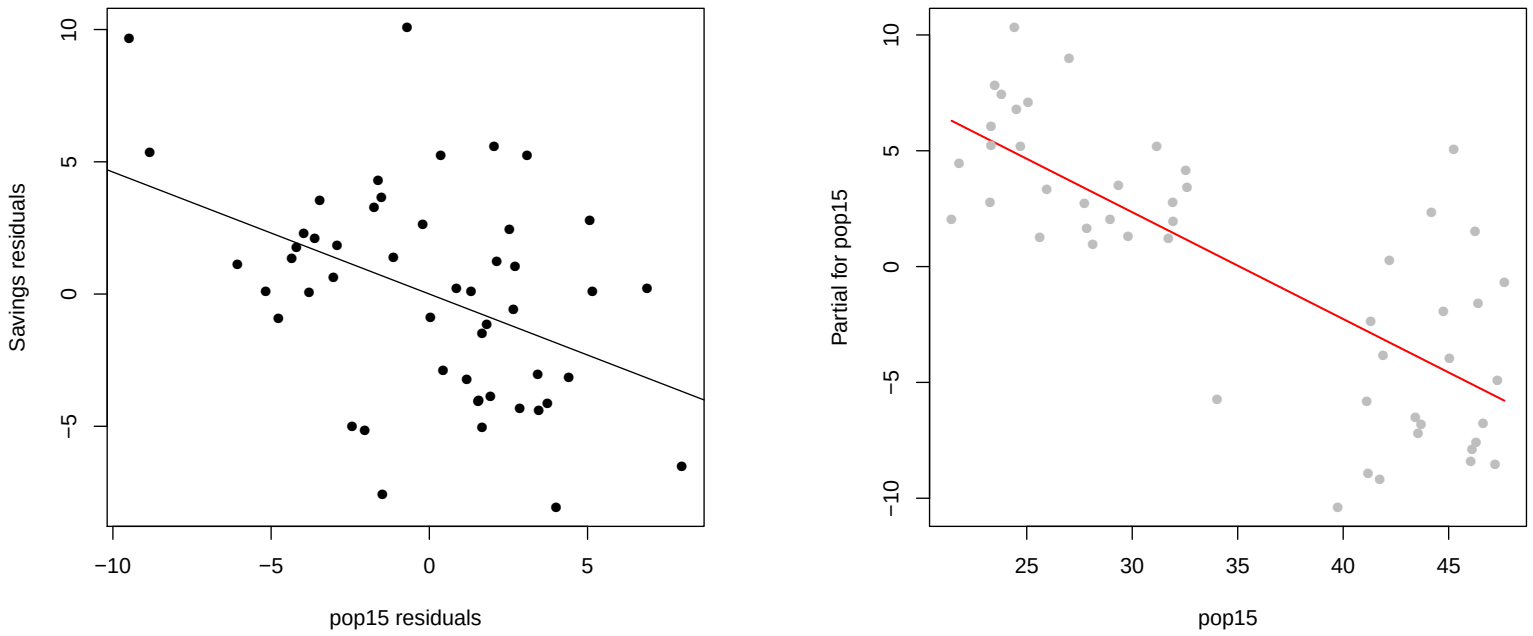


Figure: Left: Partial regression plot; Right: Partial residual plot

- Partial regression plot: the linear relationship between “sr" and “pop15" does not look very strong though; some potential outliers/influential points;
- Partial residual plot: It suggests that there may be a different relationship in the two groups.
  - pop15>35: undeveloped countries;
  - pop15<35: developed countries;

Partial Regression Plot V.S. Partial Residual Plot

- Partial regression plot: better at detecting potential outliers/influential points;
- Partial residual plot: better for non-linearity detection;

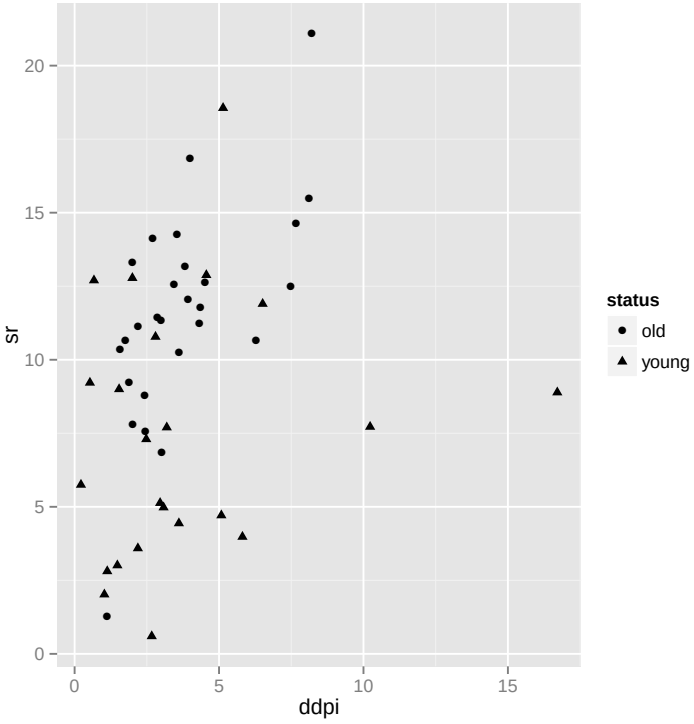
Saving Data

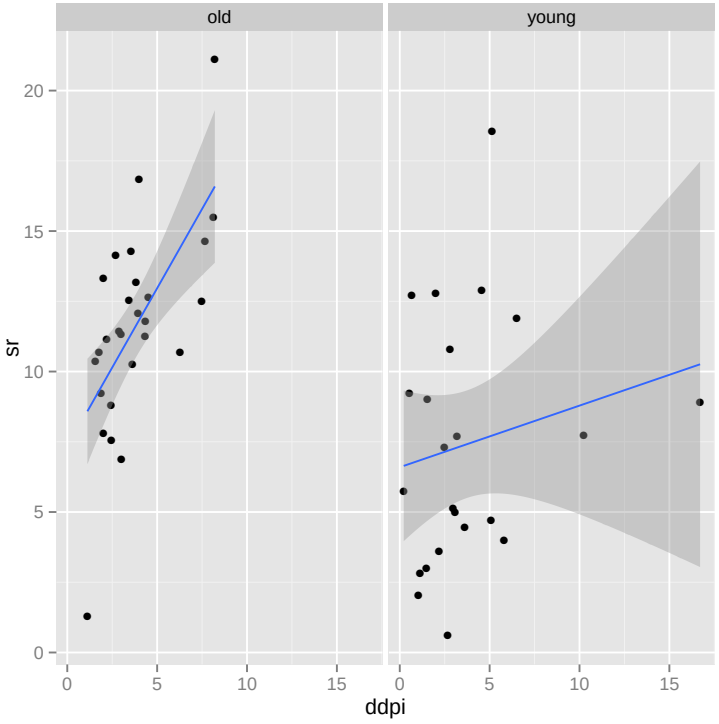
```
library(faraway)
mod1 <- lm(sr ~ pop15+pop75+dpi+ddpi,savings,subset=(pop15 > 35))
mod2 <- lm(sr ~ pop15+pop75+dpi+ddpi,savings,subset=(pop15 < 35))
> sumary(mod1)
Estimate  Std. Error t value Pr(>|t|)
(Intercept) -2.43396890 21.15502778 -0.1151   0.9097
pop15        0.27385369  0.43919097  0.6235   0.5408
pop75       -3.54847686  3.03328065 -1.1698   0.2573
dpi          0.00042076  0.00500007  0.0842   0.9339
ddpi         0.39547422  0.29010124  1.3632   0.1896
n = 23, p = 5, Residual SE = 4.45441, R-Squared = 0.16

> sumary(mod2)
Estimate  Std. Error t value Pr(>|t|)
(Intercept) 23.96179499  8.08375019  2.9642 0.007164
pop15       -0.38589762  0.19536859 -1.9752 0.060921
pop75       -1.32774213  0.92606273 -1.4337 0.165705
dpi         -0.00045881  0.00072372 -0.6340 0.532640
ddpi         0.88439443  0.29534055  2.9945 0.006679
n = 27, p = 5, Residual SE = 2.77167, R-Squared = 0.51
```

Data Visualization

```
library(faraway)
savings$status <- ifelse(savings$pop15 > 35, "young", "old")
require(ggplot2)
ggplot(savings,aes(x=ddpi,y=sr,shape=status))+geom_point()
ggplot(savings,aes(x=ddpi,y=sr))+geom_point()+facet_grid(~ status)+
stat_smooth(method="lm")
```





Recommendations

- If one can assume that all relevant predictors are at hand, and yet the model does not fit, try
  - Transformations: this includes transformations of y and predictors x's;
    - Good transformations are hard to find; There is an interesting paper: "Estimating Optimal Transformations for Multiple Regression and Correlation" (Breiman-Friedman 1985, JASA);
  - additional FUNCTIONAL TERMS such as:
    - interactions: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon$;
    - R code is $y\tilde{}x1 + x2 + x1 : x2$
    - Polynomial terms:$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \varepsilon$
    - R code is $y\tilde{}x + I(x^2) + I(x^3)$
  - Nonlinear regression model...